

Pull Back Toy Car Design Blueprint

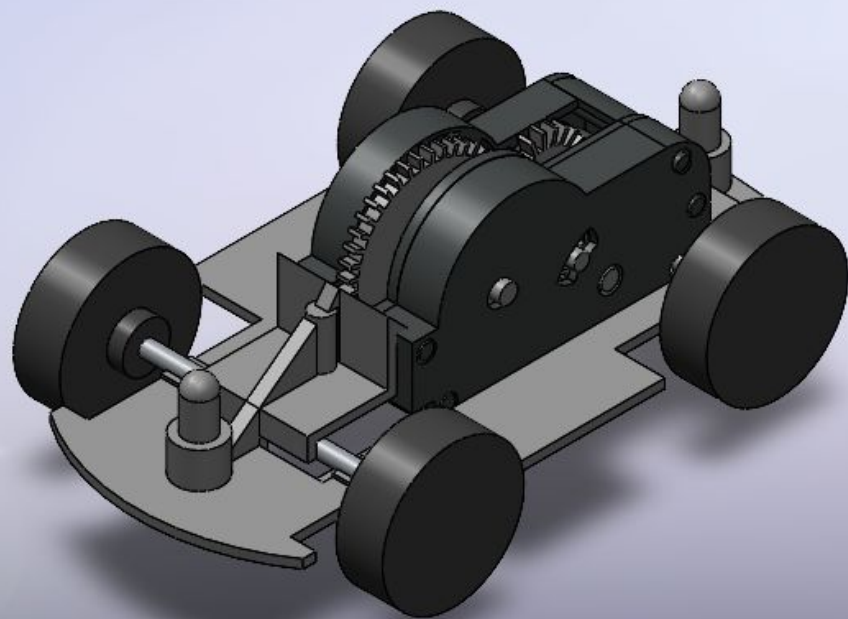
(Final Project)

24FA-DFT170-B101,B103

ZECHARIAH DIEGELMAN

DR. GARBINI

DECEMBER 10, 2024



Pull Back Toy Car

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Materials

- ABS Plastic
- AISI 1020
- Natural Rubber

Technical Data sheet (ABS Plastic) (SD3D 1)

ABS
(Acrylonitrile Butadiene Styrene)



Technical Data Sheet

Tensile Properties		
ASTM D638 - Type V		
Property	Imperial	Metric
Toughness*	1.57 ft-lb/in ²	3.3 KJ/m ²
Tensile Modulus	246000 psi	2.3 GPa
Ultimate Tensile Strength	3830 psi	26.4 MPa
Tensile Strength at Yield	5200 psi	35.9 MPa
Elongation at Yield	4%	4%
Elongation at Break	8%	8%
3D Printing Properties		
Property	Imperial	Metric
Expected Max Linear Print Speed	3.54 in/s	90 mm/s
Hardness, ASTM D2240	109.5D	109.5D
Solid Density, ASTM D792	1.17 x 10 ⁻³ slug/in ³	1.04 g/cc
Impact Properties		
Property	Imperial	Metric
Notched Izod (machined), 23 C, ASTM D256	2.6 f-lb/in	139 J/m
Gardner Impact, 23 C, ASTM D5420	22.9 ft-lb	31.1 J
Thermal Properties		
Property	Imperial	Metric
Glass Transition by DSC, ASTM E1356	221 F	105 C
Glass Transition by DMA, ASTM D792	226 F	108 C
Heat Deflection Temperature, ASTM D648	105 F	96 C
Coefficient of Thermal Expansion, ASTM E832	40 x 10 ⁻⁶ in/inR	72 x 10 ⁻⁶ m/m-K
Heat Capacity, ASTM E1269	0.400 Btu/lb/°F	1,670 J/kg-K
Thermal Conductivity, ASTM C518	1.2 Btu-in/hr/ft ² /°F	0.173 W/m-K
Available Colors		
Black, Blue, Green, Grey, Natural, Orange, Red, White, Yellow		
Suggested Uses		
ABS is one of the most cost-effective FDM printing materials. ABS is best used in mechanical applications due to it's strength, high glass transition temperature and ability to bend before breaking.		

*Toughness is not defined in ASTM D638 though can be calculated by taking the integral of the stress-strain curve collected by tensile data.

Visit www.sd3d.com/materials to learn more

Technical Data Sheet (Natural Rubber) (The Rubber Company 1)

Natural Rubber Sheet BS1154 Datasheet Product Code: RC0345-Y50

Product Description

Our Natural Rubber Sheet BS1154, as the name suggests, conforms to British standards 1154. Conforming to this requirement means that the product is certifiably good for use in highly demanding applications and environments. Our sheeting features excellent physical properties inclusive of high tensile strength and a good abrasion performance.



Available In -



Full Rolls Cut to strip Bespoke Gaskets

Technical Specification

Properties	Test Method	Values
Product		BS1154 Natural Rubber Sheet
Colour		White
Compound		NR
Hardness (Shore A)		50° ± 5°
Density		1.15 g/cm ³
Tensile Strength		17 Mpa
Elongation at Break		500%
Abrasion Resistance		125 mm ³
Working Temperature Range		-20 °C to +70 °C



The information contained on this product information sheet is to be used as guidance. The advice is given in good faith and does not constitute any guarantee or recommendation for suitability. The Rubber Company cannot be held liable for any damage caused by incorrect installation. We hereby reserve the right to change the technical information hereafter without notification or prior agreement.

Technical Data Sheet (Natural Rubber) (The Rubber Company 2)



Experts in
Rubber

Standard Sizes

Stock Roll Sizes	Weight (Kg/m ²)
3mm x 1.4m x 10m	3.45
5mm x 1.4m x 10m	5.75
6mm x 1.4m x 10m	6.90



The information contained on this product information sheet is to be used as guidance. The advice is given in good faith and does not constitute any guarantee or recommendation for suitability. The Rubber Company cannot be held liable for any damage caused by incorrect installation. We hereby reserve the right to change the technical information hereafter without notification or prior agreement.

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Technical Data Sheet (AISI 1020) (Smiths 1)

1020 (UNS G10200)

Technical Datasheet



Aerospace Carbon Steel Tube

Service. Quality. Value.

Typical Applications

- Shafts
- Chains
- Lightly stressed gears
- Pins
- Hard wearing surfaces
- Case Hardened components where core strength is essential

Product Description

A plain carbon steel with moderate strength and limited hardenability. Finds applications in general engineering and some motorsports where required performance requirements are relatively low and strength to weight ratios less important. The material has a good combination of strength and ductility and can be hardened or carburised. 1020 is commonly used in the case hardened condition.

Key features:

- Good combination of strength and ductility
- Limited hardenability
- Good machinability
- Easily welded
- For simple structural applications

Machinability & Weldability

Good machinability (65%). Can be easily welded using traditional methods.

Availability

Tube

Chemical Composition (weight %)

	Mn	C	S	P	Fe					
Min	0.30	0.18			Bal					
Max	0.60	0.23	0.05	0.04	Bal					

Mechanical Properties (cold rolled condition)

Tensile strength	420	MPa
Yield strength	350	MPa
Modulus of elasticity	205	GPa
Shear modulus (typical for steel)	80	GPa
Poisson's ratio	0.29	
Elongation at break (in 50 mm)	15	%
Hardness, Brinell	121	
Hardness, Knoop (converted from Brinell hardness)	140	
Hardness, Rockwell B (converted from Brinell hardness)	68	
Hardness, Vickers (converted from Brinell hardness)	126	
Machinability (based on AISI 1212 steel, as 100 machinability)	65	

Technical Assistance

Our knowledgeable staff backed up by our resident team of qualified metallurgists and engineers, will be pleased to assist further on any technical topic.

UK Service Centres:

Smiths Belfast	02895 908 897	Smiths Leeds	0113 307 5167
Smiths Biggleswade	01767 664 794	Smiths London	020 7241 2430
Smiths Birmingham	0121 728 4940	Smiths Manchester	0161 794 8550
Smiths Bristol	0117 971 2800	Smiths Norwich	01603 789 678
Smiths Chalfont	01295 466 664	Smiths Nottingham	0115 925 4801
Smiths Gateshead	0191 489 5428	Smiths Redruth	01209 315 512
Smiths Hornham	01463 261 981	Smiths Warwick	01202 826 347

Quality & Testing:





0845 527 3331
www.smithmetal.com info@smithmetal.com

All information in our data sheet is based on appropriate testing and is stated to the best of our knowledge and belief. It is presented apart from contractual obligations and does not constitute any guarantee of properties or of processing or application possibilities in individual cases. Our warranties and liabilities are stated exclusively in our terms of trading.
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Technical Data Sheet (Very Low Density PE) (Laminated Plastics 1)

TECHNICAL DATA SHEET Low Density Polyethylene (LDPE)

This extruded material offers good corrosion resistance and low moisture permeability. It can be used in applications where corrosion resistance is important, but stiffness, high temperature, and structural strength are not. A highly flexible product, LDPE is used widely in orthopedic products, or where mobility without stress fatigue is desired. LDPE is also frequently used in consumer packaging, bags, bottles, and liners.

TYPICAL PROPERTIES OF POLYETHYLENE

ASTM or UL test	Property	LDPE	HDPE	UHMW
PHYSICAL				
D792	Density (lb/in ³) (g/cm ³)	0.033 0.92	0.035 0.96	0.034 0.93
D570	Water Absorption, 24 hrs (%)	<0.01	<0.01	<0.01
MECHANICAL				
D638	Tensile Strength (psi) at 72°F	1,400	4,600	5,800
D638	Tensile Strength (psi) at 150°F	400	400	400
D638	Tensile Modulus (psi)	57,000	200,000	80,000
D638	Tensile Elongation at Break (%)	100	400	300
D790	Flexural Strength at Yield (psi)	1,500	4,600	3,500
D790	Flexural Modulus (psi)	29,000	174,000	88,000
D695	Compressive Strength (psi)	1,400	4,600	3,000
D695	Compressive Modulus (psi)	54,000	100,000	80,000
D732	Shear Strength (psi)	-	-	3,000
D785	Hardness, Shore D	D45	D69	D62-D66
D256	IZOD Notched Impact (ft-lb/in)	No Break	1.3	No Break
THERMAL				
D696	Coefficient of Linear Thermal Expansion (x 10 ⁻⁵ in./in./°F)	-	6	11
D648	Heat Deflection Temp (°F / °C) at 66 psi at 264 psi	120 / 48 116 / 46	170 / 76 176 / 80	203 / 95 180 / 82
D3418	Approx. Melting Temperature (°F / °C)	244 / 118	260 / 125	275 / 136
-	Max Operating Temp (°F / °C)	160 / 71	180 / 82	180 / 82
C177	Thermal Conductivity (BTU-in/ft ² -hr-°F) (x 10 ⁻⁸ cal/cm-sec-°C)	-	-	2.84 10.0
UL94	Flammability Rating	HB	HB<	HB
ELECTRICAL				
D149	Dielectric Strength (V/mil) short time, 1/8" thick	460-700	450-500	2300
D150	Dielectric Constant at 1 MHz	2.25- 2.30	2.30- 2.35	2.30- 2.35
D150	Dissipation Factor at 1 kHz	0.0002	0.0002	0.0005
D257	Surface Resistivity (ohm/square) at 50% RH	> 10 ¹³	> 10 ¹³	> 10 ¹³
D495	Arc Resistance (sec)	135-160	200-250	250-350

Benefits

- Lightweight
- Formable
- Impact Resistant
- Good electrical properties
- Easily cleaned
- Easily Fabricated

Applications

- Chemical resistant tank and containers
- Food storage containers
- Laboratory equipment
- Corrosion resistant work surfaces
- Vacuum formed end caps and tops
- Moisture barrier

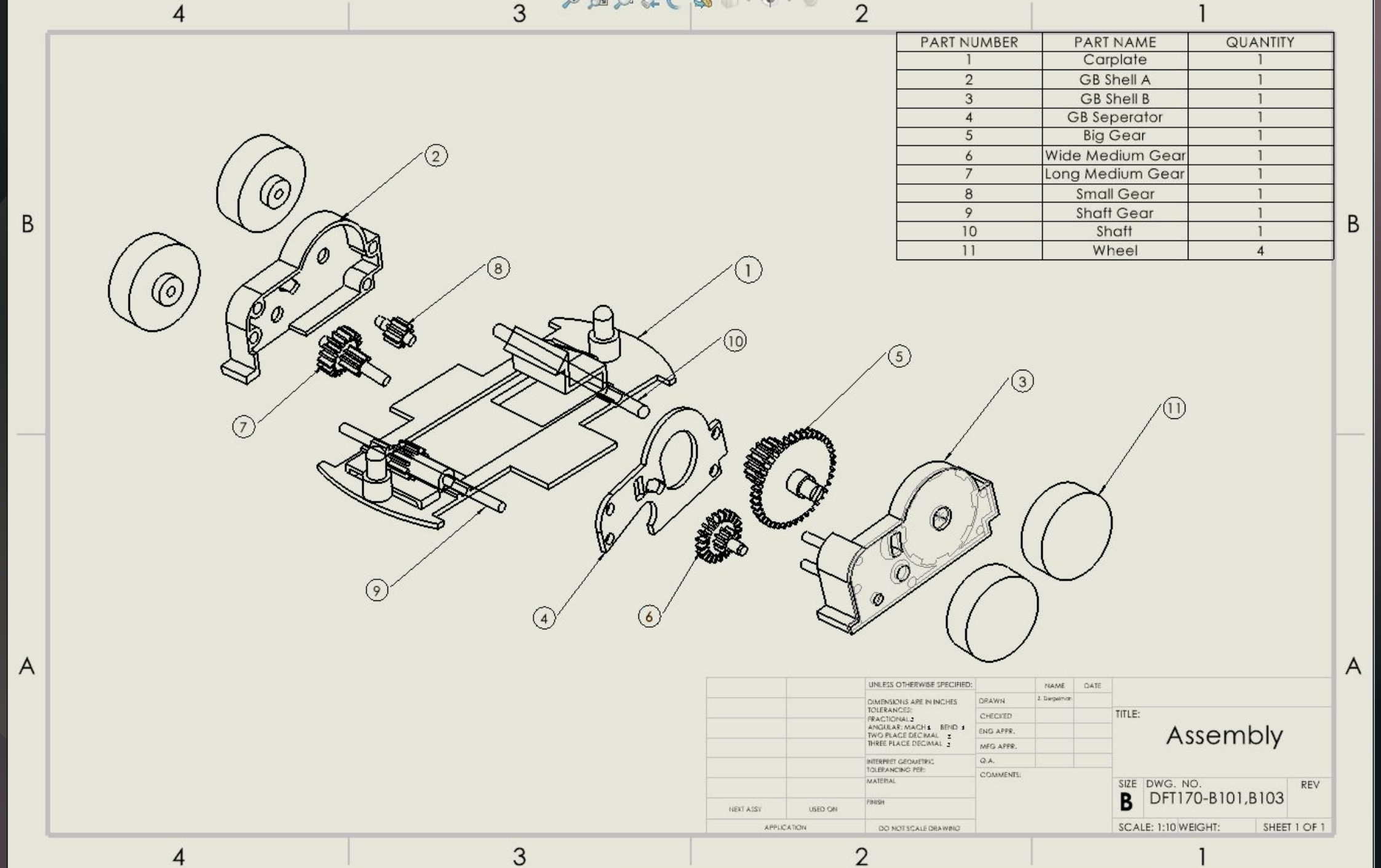
SHAPES AVAILABLE



NOTE: The information contained herein are typical values intended for reference and comparison purposes only. They should NOT be used as a basis for design specifications or quality control. Contact us for manufacturers' complete material property datasheets. All values at 73°F (23°C) unless otherwise noted.

Drawing Overview

- This overview includes exploded drawing with itemized part list.



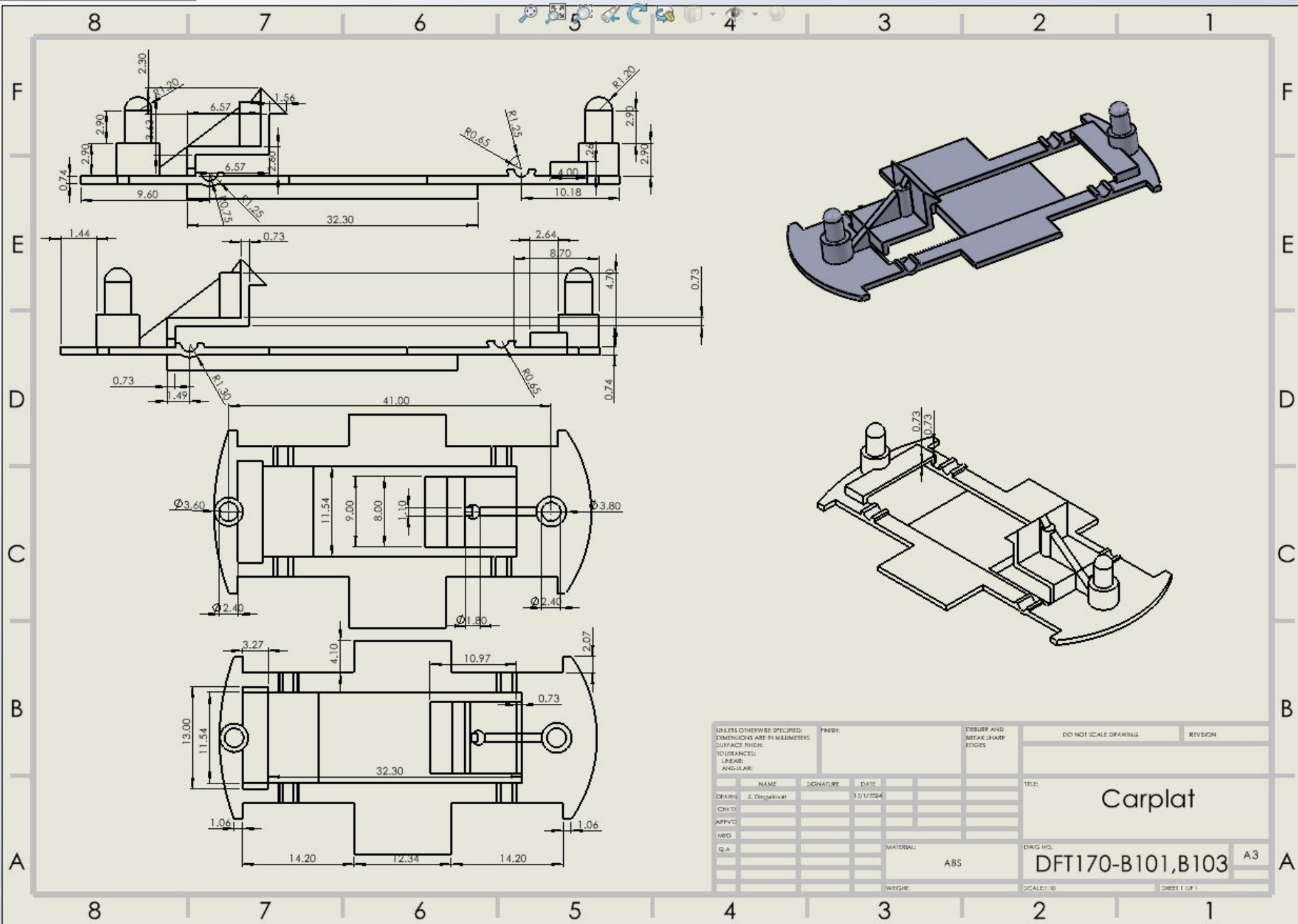
PART NUMBER	PART NAME	QUANTITY
1	Carplate	1
2	GB Shell A	1
3	GB Shell B	1
4	GB Seperator	1
5	Big Gear	1
6	Wide Medium Gear	1
7	Long Medium Gear	1
8	Small Gear	1
9	Shaft Gear	1
10	Shaft	1
11	Wheel	4

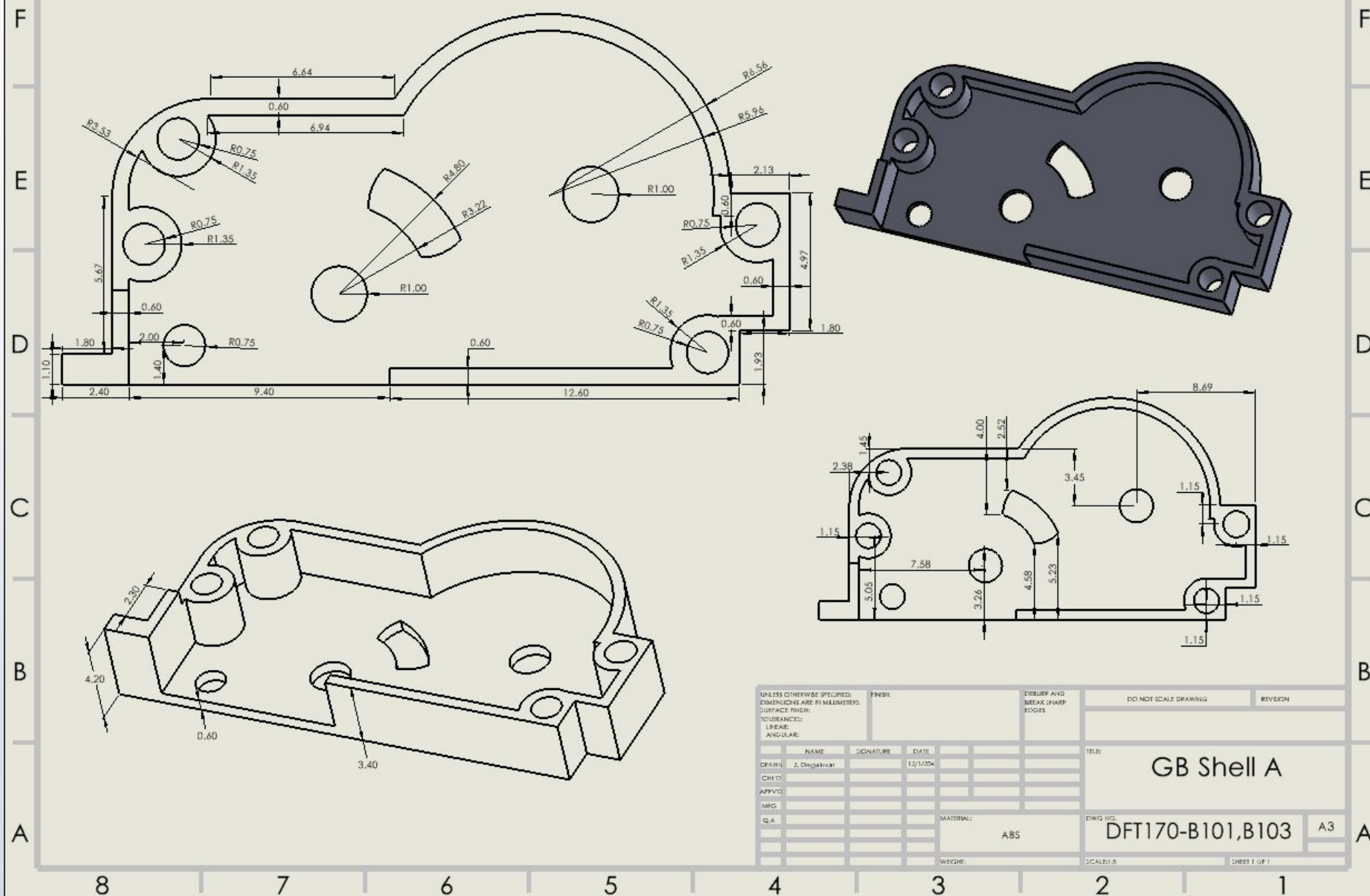
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DIMENSIONS ARE IN INCHES		DRAWN	J. Siegelman
TOLERANCES:		CHECKED	
FRACTIONAL 1/16		ENG APPR.	
ANGULAR: MACH 1/2 BEND 1/2		MFG APPR.	
TWO PLACE DECIMAL .01		Q.A.	
THREE PLACE DECIMAL .005		COMMENTS:	
INTERPRET GEOMETRIC TOLERANCING PER:			
MATERIAL:			
NEXT ASSY	USED ON	FINISH	
APPLICATION		DO NOT SCALE DRAWING	

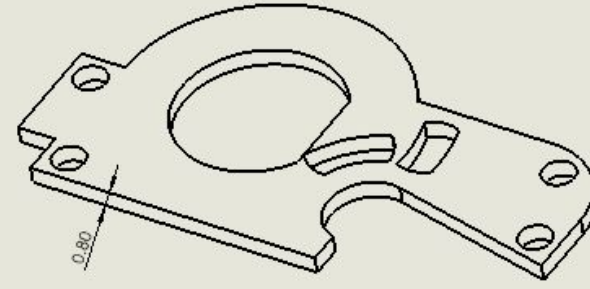
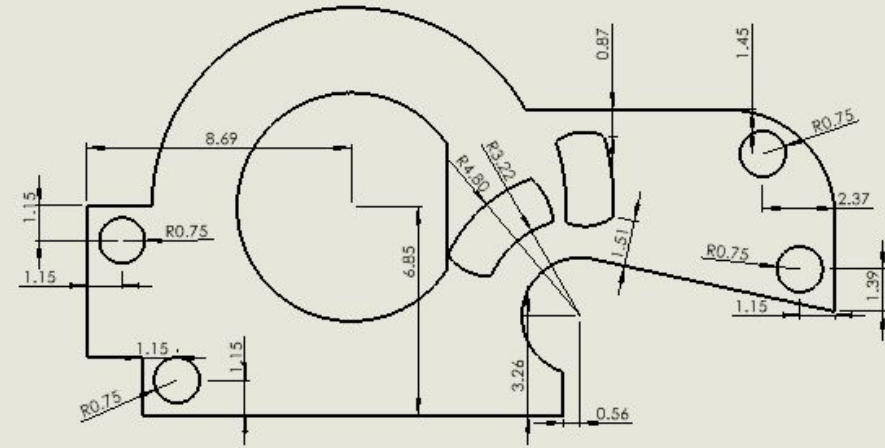
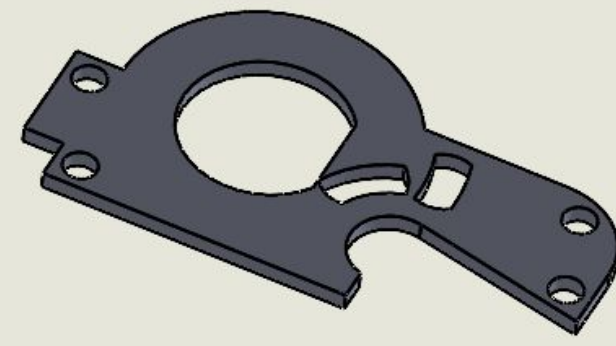
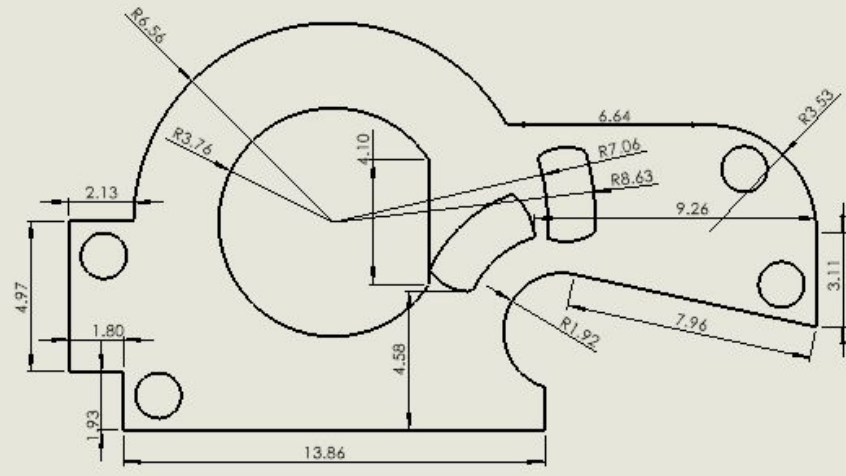
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Main Assembly Bodies

- Carplat
- GB Shell A
- GB Shell B
- GB Separator



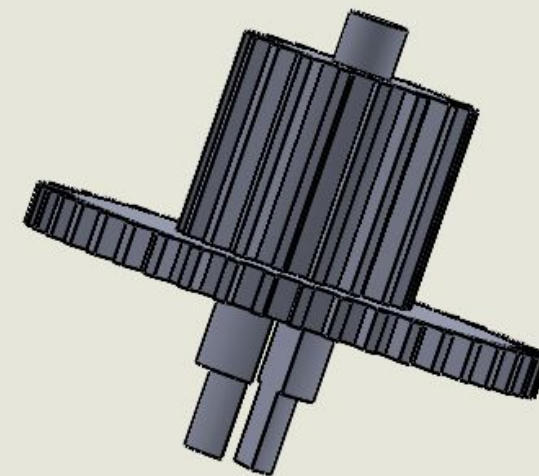
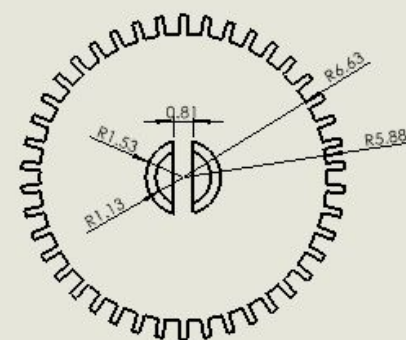
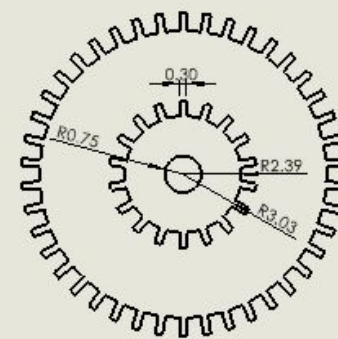
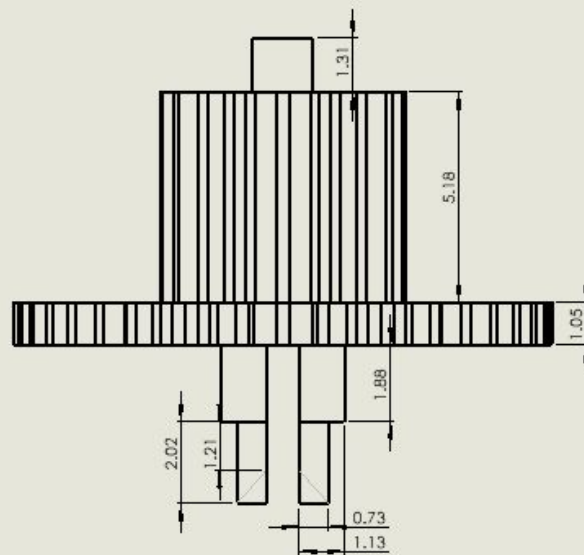




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NAME	SIGNATURE	DATE		TITLE:	
DRAWN: J. Dispatrican		12/12/2024		GB Seperator	
CHECKED:				DWG NO:	
APPROVED:				DFT170-B101,B103	
MFG:				A3	
QA:			MATERIAL: ABS	SHEET 1 OF 1	
			WEIGHT:		

Gear System

- Big Gear
- Wide Medium Gear
- Long Medium Gear
- Small Gear
- Shaft Gear

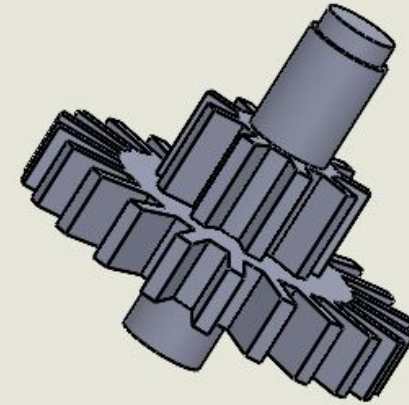
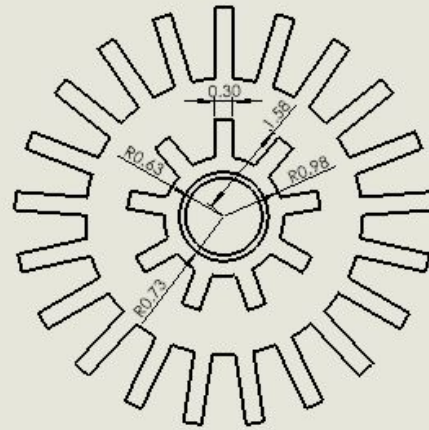
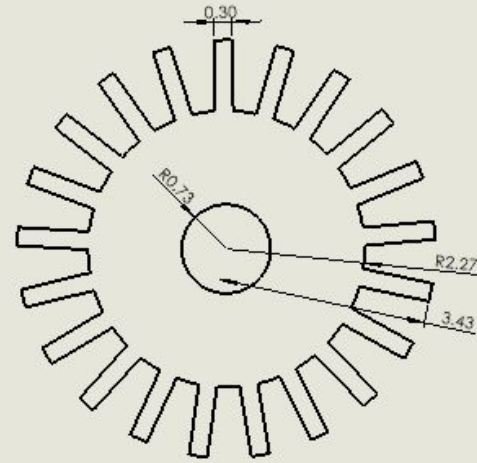
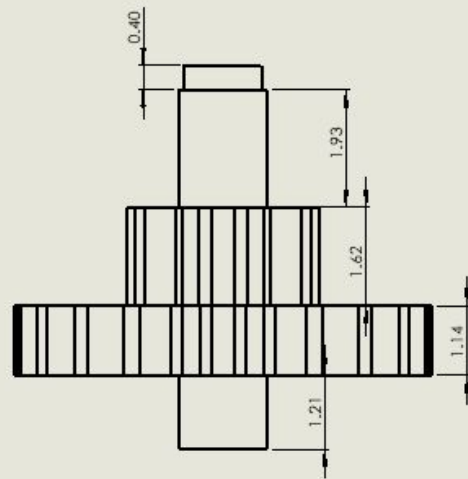


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CHECKED					
APPROVED					
MRG					
Q.A.					
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WEIGHT:			SCALE: 1:1		SHEET 1 OF 1

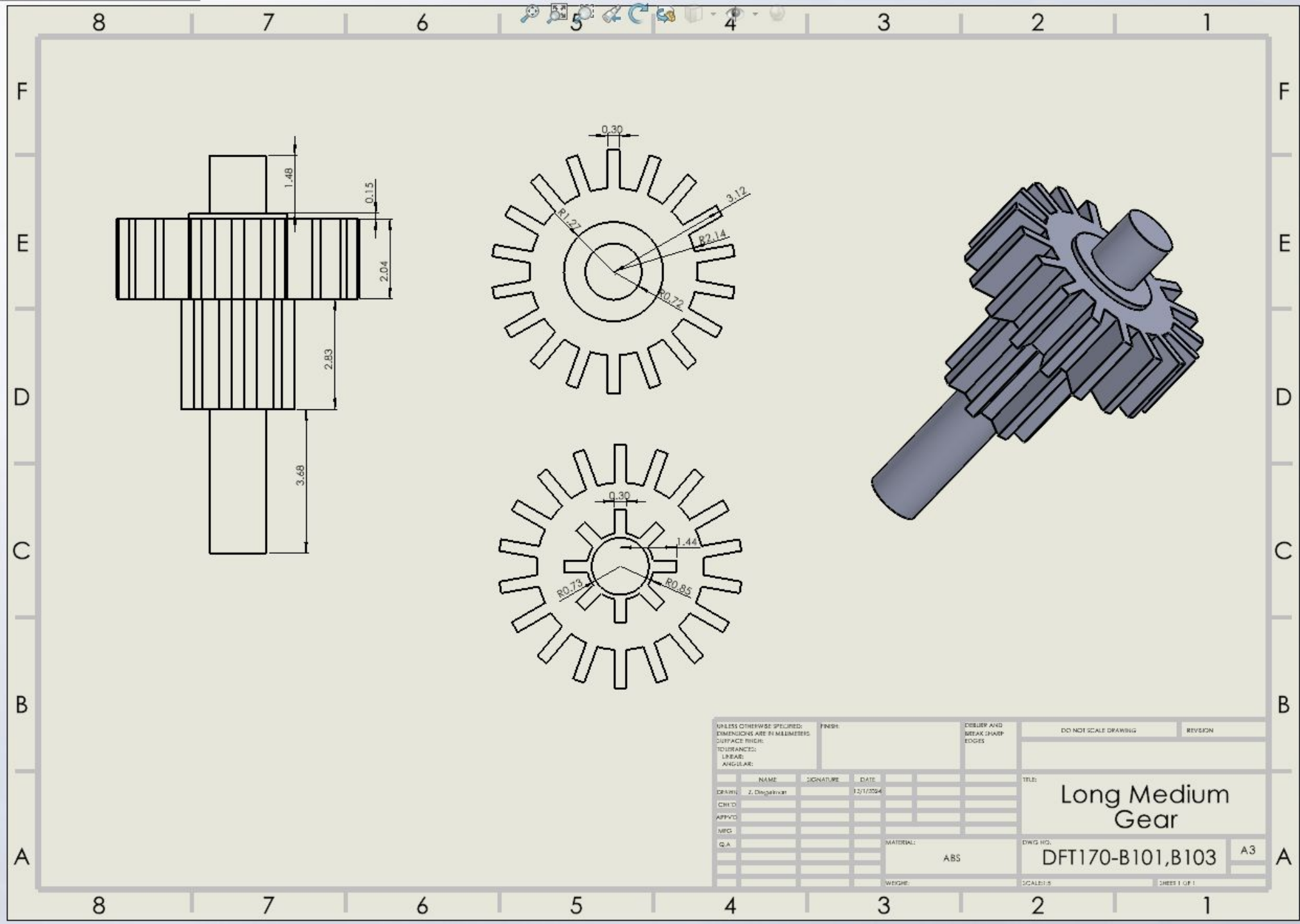
Large Gear

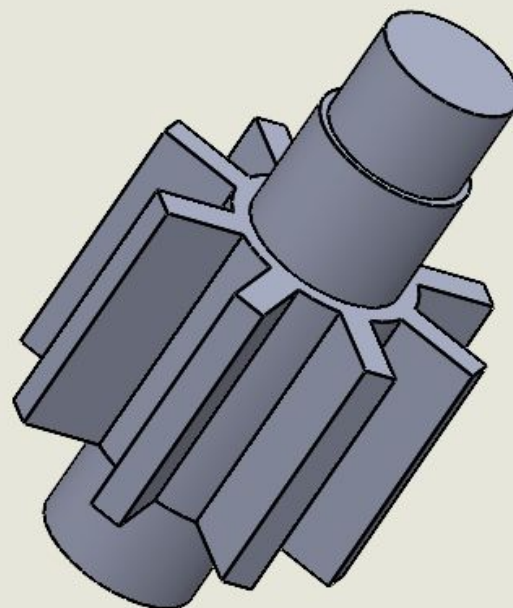
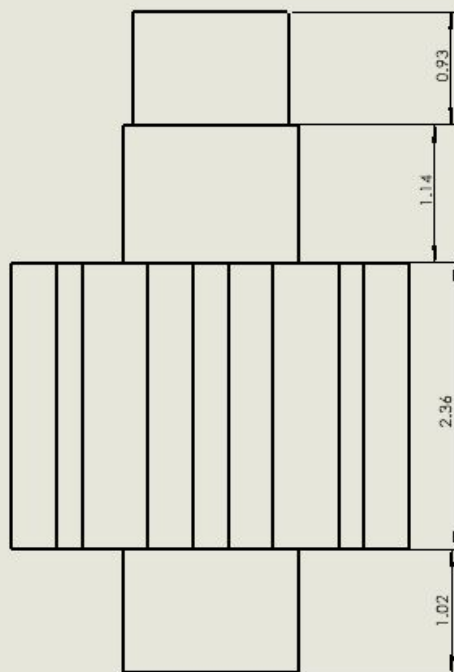
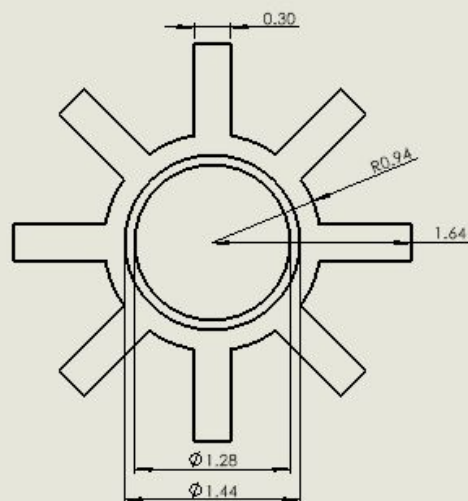
DFT170-B101,B103

A3

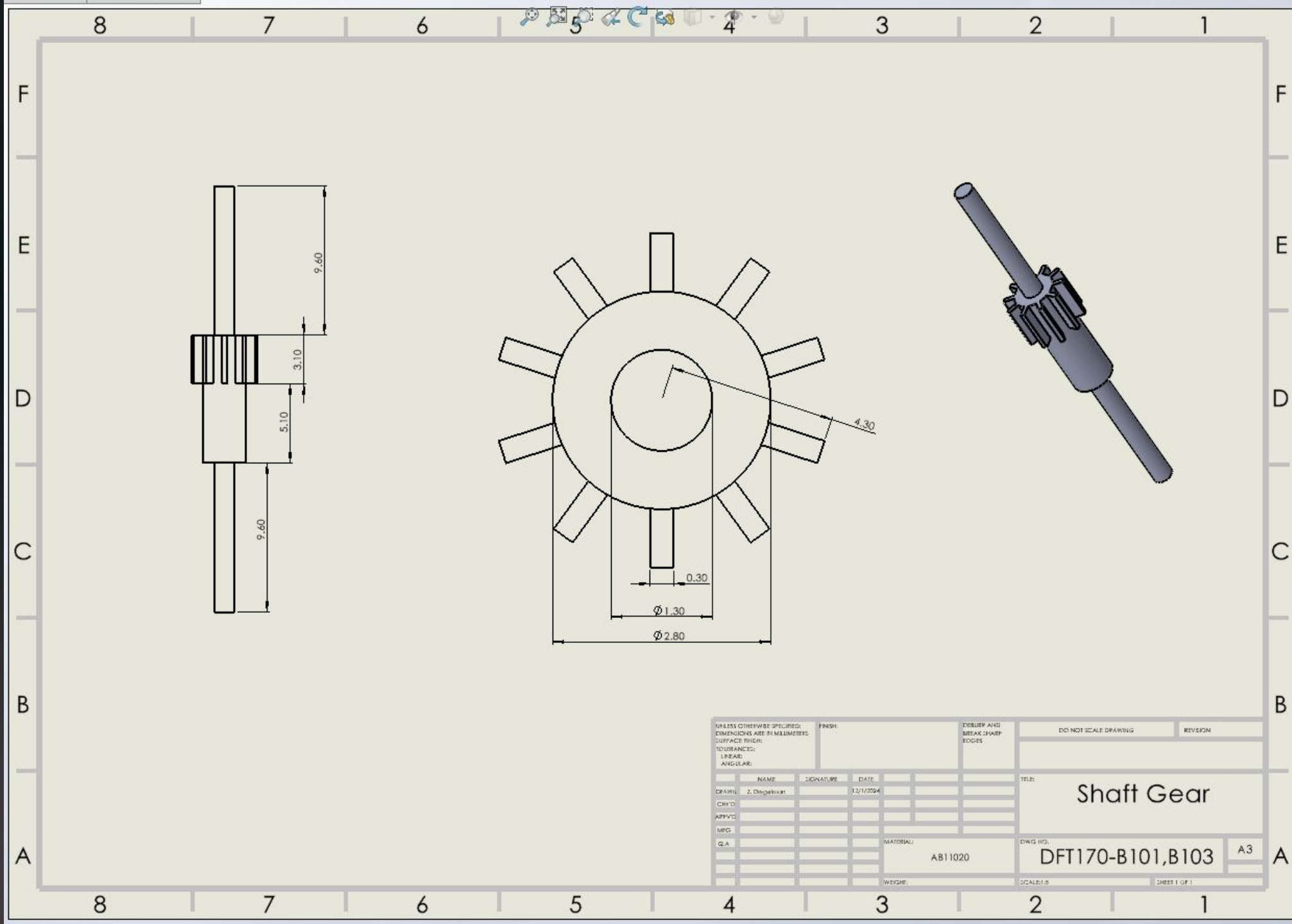


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SURFACE FINISH:									
TOLERANCES:									
LINEAR:									
ANGULAR:									
NAME	SIGNATURE	DATE				TITLE:			
DESIGN	Z. Deygudov	12/1/2024				Wide Medium Gear			
CHK'D						DWG NO.			
APPROV						DFT170-B101,B103			
WFO						A3			
QA						SCALE: 1:1			
				MATERIAL:					
				A85					
				WEIGHT:					





UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: LINEAR: ANGULAR:		DATE: 3/1/2024	DESIGNER AND BREAK SHARP EDGES	DO NOT SCALE DRAWING	REVISION
NAME	SIGNATURE	DATE	TELE:	Small Gear	
DRAWN	J. Delgadillo	3/1/2024			
CHECKED					
APPROVED					
REVIEW					
QA					
MATERIAL: ABS			DWG NO. DFT170-B101,B103		A3
WEIGHT:			SCALE: 1:2		SHEET 1 OF 1



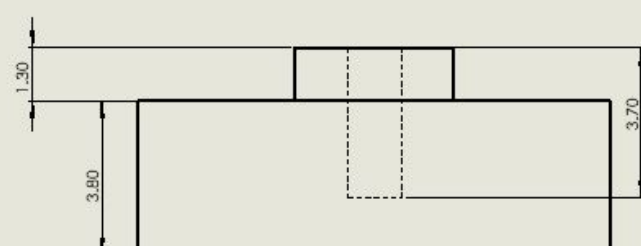
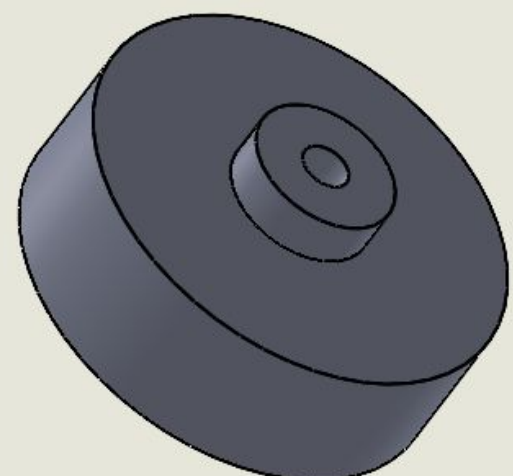
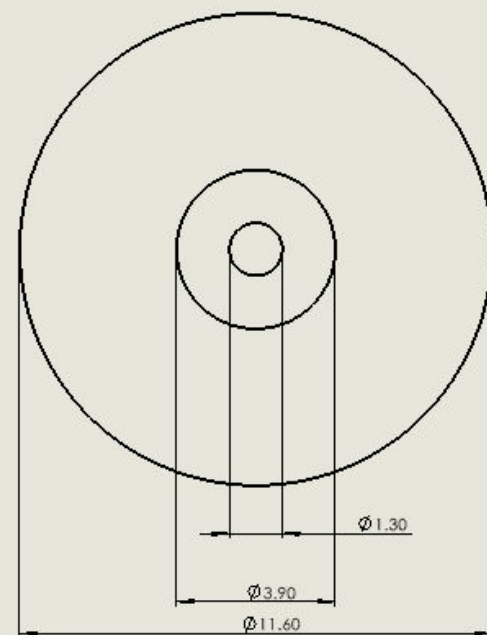
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APPROVED:					
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Q.A.					
MATERIAL:			DWTG NO.		
AB11020			DFT170-B101,B103		
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Shaft Gear

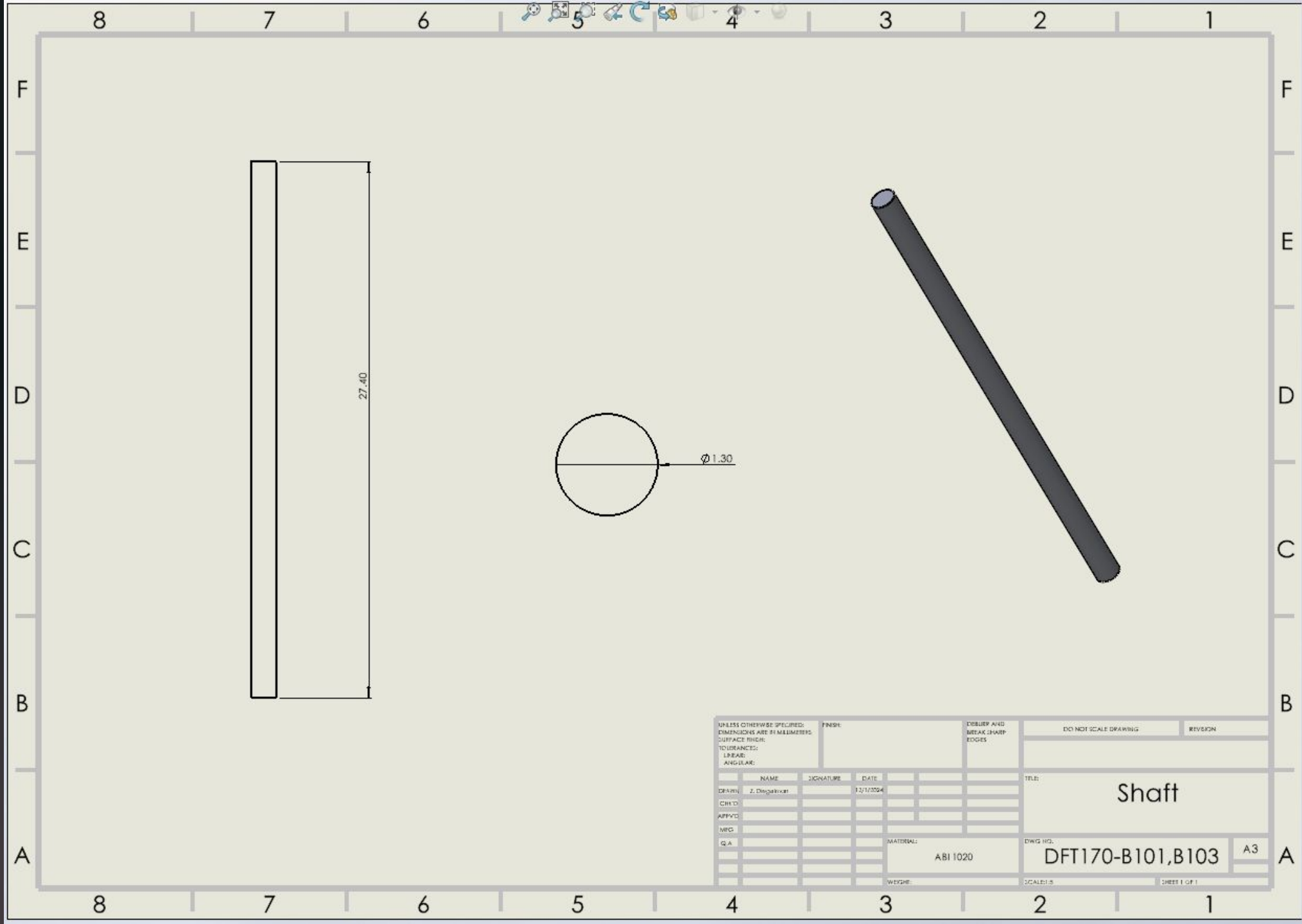
A3

Motion System

- Wheel
- Shaft



UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: LINEAR: ANGULAR:			FINISH:		DEBurr AND BREAK SHARP EDGES		DO NOT SCALE DRAWING		REVISION	
NAME			SIGNATURE		DATE		TITLE:		Wheel	
DRAWN: J. Desai					12/1/2024					
CHECKED:										
APPROVED:										
MATERIAL:					ABS		DWG NO.:		DFT170-B101,B103	
WEIGHT:							SCALE:		1:1	
									SHEET 1 OF 1	



New parts for Car Plane Design

- Aileron
- Wings
- Wing Support
- Top Support
- Support Beam Front
- Support Beam Back
- Lever Mechanism
- Lever Handle
- Bevel Gear Small
- Bevel Gear Large
- Double Gear Long
- Gear Support Large
- Gear Support Small
- Propeller with Spinning Hook
- Hook with Gear
- Prop Rubber Band

New Material

The new materials used is the 5052 H32 Aluminum, which will be used for the gear supports to ensure durability and stability, even in the event of "crash landings." This aluminum is both strong enough to hold the components securely and light enough to optimize airtime. PETG will be utilized for the wings and wing support due to its lightweight properties compared to most metals and its superior strength over many plastics. Additionally, a tungsten alloy, the heaviest material in the design, will be incorporated to secure all components together at the top support beam, ensuring overall structural integrity.

5052 H32

Aluminum

Coremark

Metals



Aluminium Alloy Data Sheet 5052

Specified Properties

These properties are specified for flat rolled product (plate, sheet and coil) in ASTM B209M. Similar but not necessarily identical properties are specified for other products such as tube and bar in their respective specifications.

Composition Specification (%) (single values are maxima except as noted)

Alloy	Si	Fe	Cu	Mn	Mg	Cr	Zn	Ti	Others	
									Each	Total
5052	0.25	0.40	0.10	0.10	2.2-2.8	0.15-0.35	0.10	-	0.05	0.15

Mechanical Property Specification (single values are minima except as noted)

Alloy & Temper	Tensile Strength (MPa)	Yield Strength 0.2% Proof (MPa) min	Elongation (% in 50mm) minimum for sheet or plate thicknesses shown				
			0.15-0.32mm	0.33-0.63mm	0.64-1.20mm	1.21-6.30mm	6.31-80.0mm
5052-O	170 - 215	65	13	15	17	19	18
5052-H32	215 - 265	160	-	4	5	7	11
5052-H34	235 - 285	180	3	3	4	6	10
5052-H36	255 - 305	200	2	3	4	4*	-
5052-H38	270 min	220	2	3	4	4*	-

Temper H22 and H24 etc may not meet maximum tensile strength and minimum yield strength limits, but if supplied in place of ordered H32 or H34 all limits must be complied with.

* properties for H36 only specified up to 4.0mm, and for H38 up to 3.2mm thick.

Specialist tempers such as F, H116, H112 and H141 are also possible in 5052 - refer to standards for details.

Physical Properties

(typical values)

Alloy	Density (kg/m ³)	Elastic Modulus (GPa)	Mean Coefficient of Thermal Expansion 20-100°C (µm/m/°C)	Thermal Conductivity at 25°C (W/m.K)	Electrical Conductivity MS/m at 20°C		Electrical Resistivity (nΩ.m)
					Equal Volume	Equal Mass	
5052	2680	70	23.8	138	20	67	50

Grade Specification Comparison

Alloy	UNS	ISO	BS	DIN	
	No			No	Name
5052	A95052	AlMg2.5	L80, L81	3.3523	AlMg2.5

These comparisons are approximate only. The list is intended as a comparison of functionally similar materials **not** as a schedule of contractual equivalents. If exact equivalents are needed original specifications must be consulted.

Possible Alternative Alloys

Alloy	Why it might be chosen instead of 5052
3003	Brighter appearance required, particularly when considering treadplate.
5005	Bright (decorative) anodising finish is required.
5083	Higher strength or improved corrosion resistance required, particularly for ship hull applications.

PETG Technical Data Sheet (TDS)



3D printing solutions for high performance materials

PETG Technical Data Sheet (TDS)

Polyethylene Terephthalate Glycol-modified (PETG) is a tough and durable materials and it is easy to use. The strength of it allow to be a suitable material for food packaging, the chemical non-reactive characteristic allow it to be the most valuable material in medical field, normally used for orthopedic and prosthetic device.

IEMAI 3D high performance PETG filament is based on FFF/FDM technology, with a commonly used diameter of 1.75 mm, 210-235°C printing temperature, 50-80 °C bed temperature, having excellent interlayer adhesion which greatly improve the strength and shock resistance of the prototype.

PETG is a very waterproof material which make it become an excellent choice for outdoor activity. It also has excellent chemical resistance, which can be well-used in both acidic and alkaline environment, PETG also have strong impact resistance, which make it be the substitute material for PMMA and PC.

Physical	Condition	Test Method	Typical Value
Density		ASTM D792	1.24 g/cm ³ at 21 °C
Bulk Density			0.73 g/cm ³
Intrinsic Viscosity		ISO 1628-5	0.80 dl/g
Water Absorption		ASTM D570	0.12%
Colour	b*	ASTM D6290	≤ 1
	L*	ASTM D6290	≥64

Mechanical	Condition	Test Method	Typical Value
Tensile Modulus		ISO 527-2	3000 MPa
Tensile Yield Stress		ISO 527-2	53 MPa
Elongation at Yield		ISO 527-2	4%
Tensile Strength		ISO 527-2	53 MPa
Elongation at Stress		ISO 527-2	4%
Stress at Break		ISO 527-2	19 MPa
Nominal Elongation at Break		ISO 527-2	31%
Flexural Modulus		ISO 178	2040 MPa
Flexural Stress		ISO 178	171 MPa
Deflection at Flexural Strength		ISO 178	8.6 mm

Tungsten Alloy

TUNGSTEN

Alpha Resources LLC

Chemwatch: 32371

Version No: 5.1.1.1

Safety Data Sheet according to OSHA HazCom Standard (2012) requirements

Chemwatch Hazard Alert Code: 2

Issue Date: 09/18/2018

Print Date: 04/11/2019

S.GHS,USA,EN

SECTION 1 IDENTIFICATION

Product Identifier

Product name	TUNGSTEN
Chemical Name	tungsten
Synonyms	W; wolfram
Chemical formula	W
Other means of identification	Not Available
CAS number	7440-33-7

Recommended use of the chemical and restrictions on use

Relevant identified uses	Lamp filaments, machine tools and metal to glass seals.
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Name, address, and telephone number of the chemical manufacturer, importer, or other responsible party

Registered company name	Alpha Resources LLC
Address	3090 Johnson Road MI 49127 United States
Telephone	(800)833-3083
Fax	(269)465-3629
Website	http://www.alpharesources.com
Email	Not Available

Emergency phone number

Association / Organisation	Not Available
Emergency telephone numbers	Not Available
Other emergency telephone numbers	Not Available

SECTION 2 HAZARD(S) IDENTIFICATION

Classification of the substance or mixture

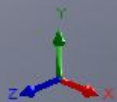
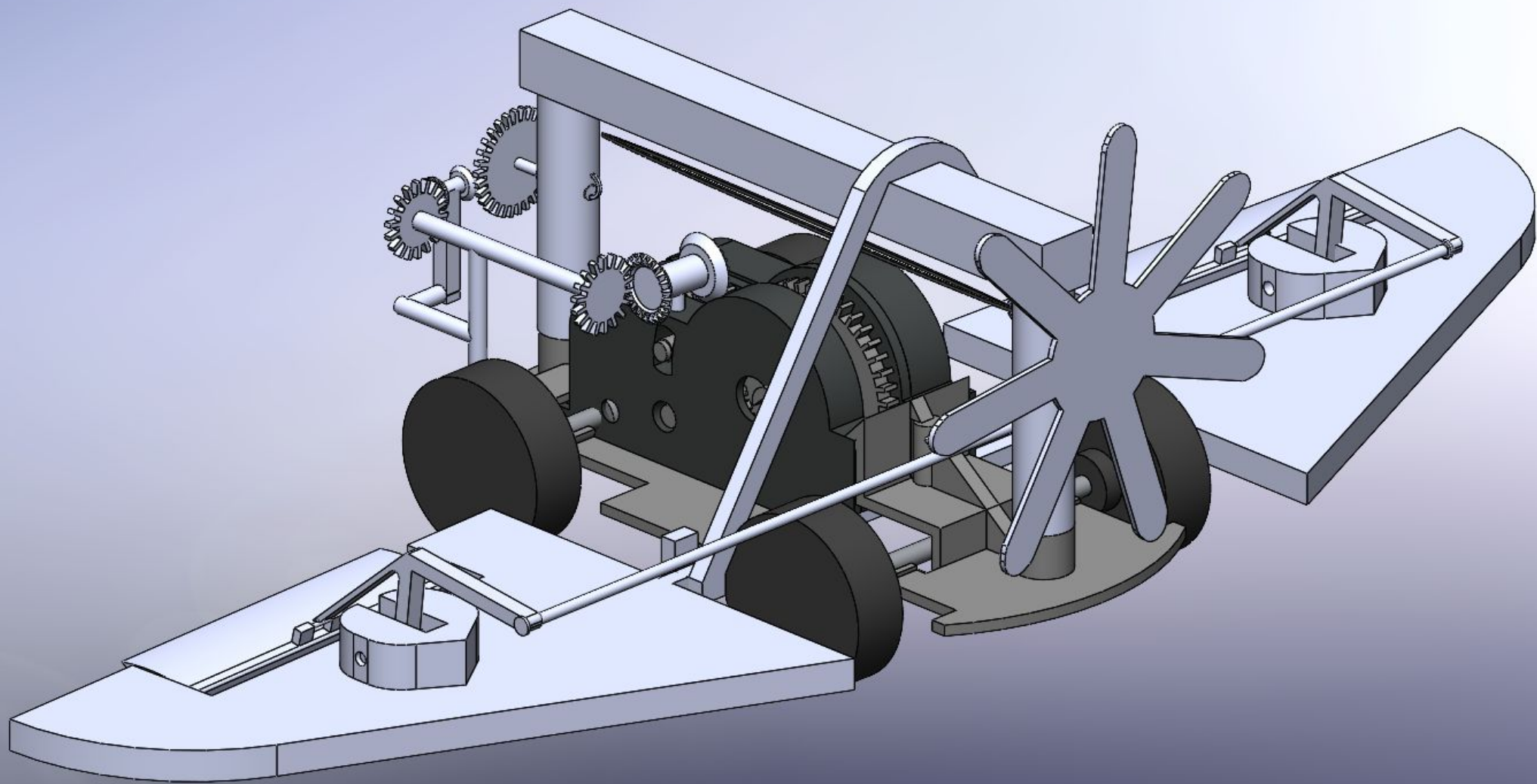
NFPA 704 diamond

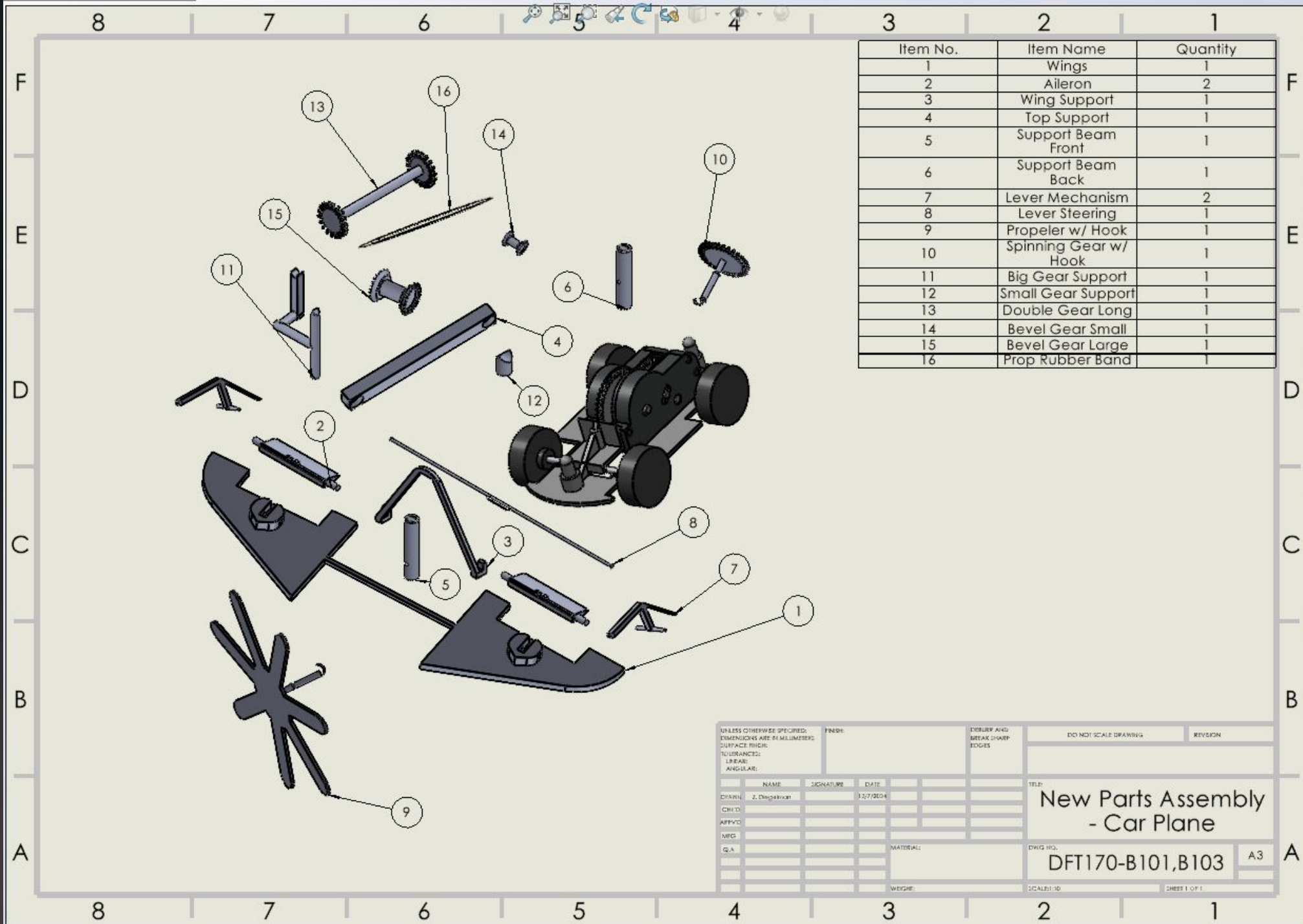


Note: The hazard category numbers found in GHS classification in section 2 of this SDSs are NOT to be used to fill in the NFPA 704 diamond. Blue = Health Red = Fire Yellow = Reactivity White = Special (Oxidizer or water reactive substances)

Classification	Eye Irritation Category 2B
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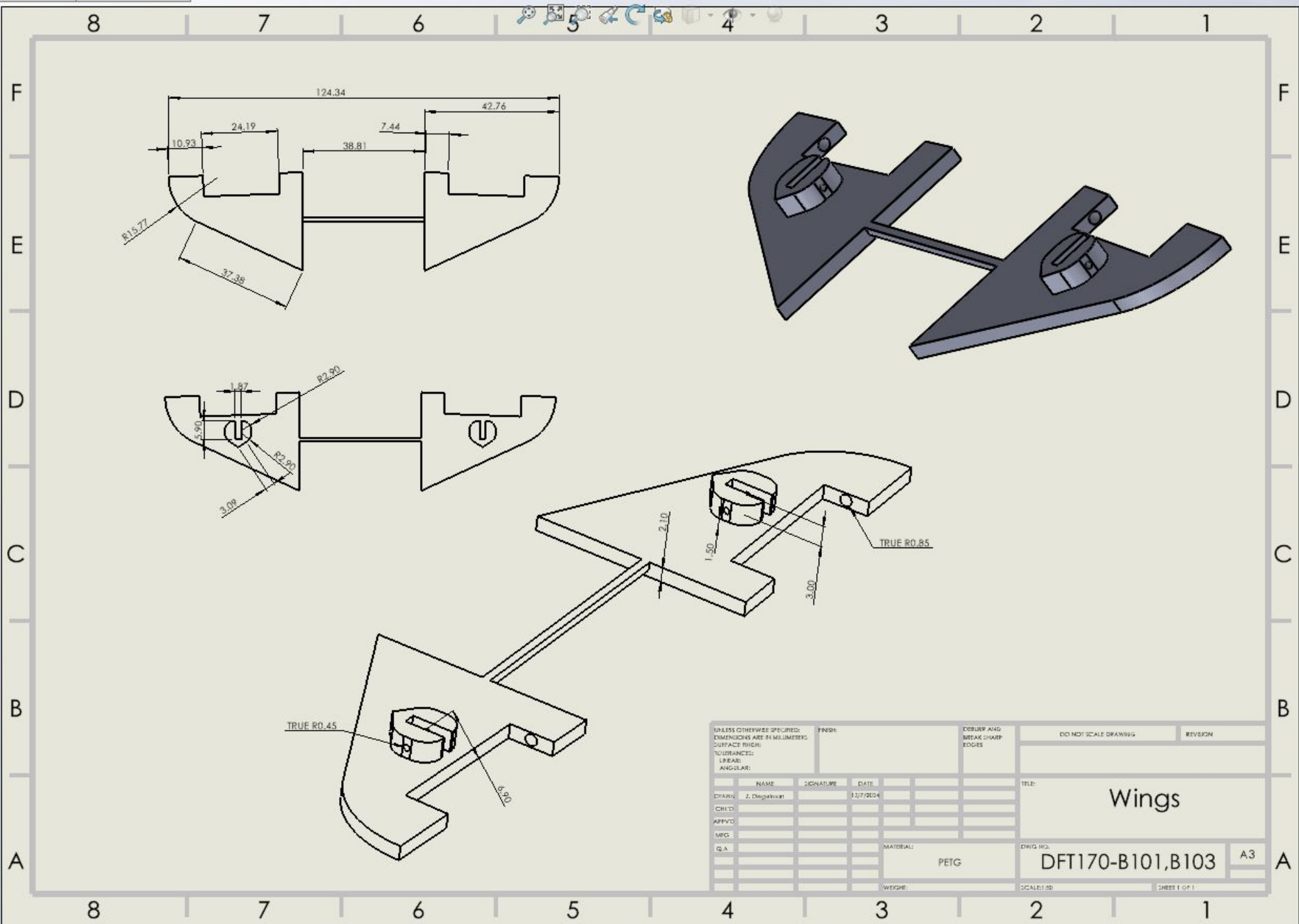
Label elements

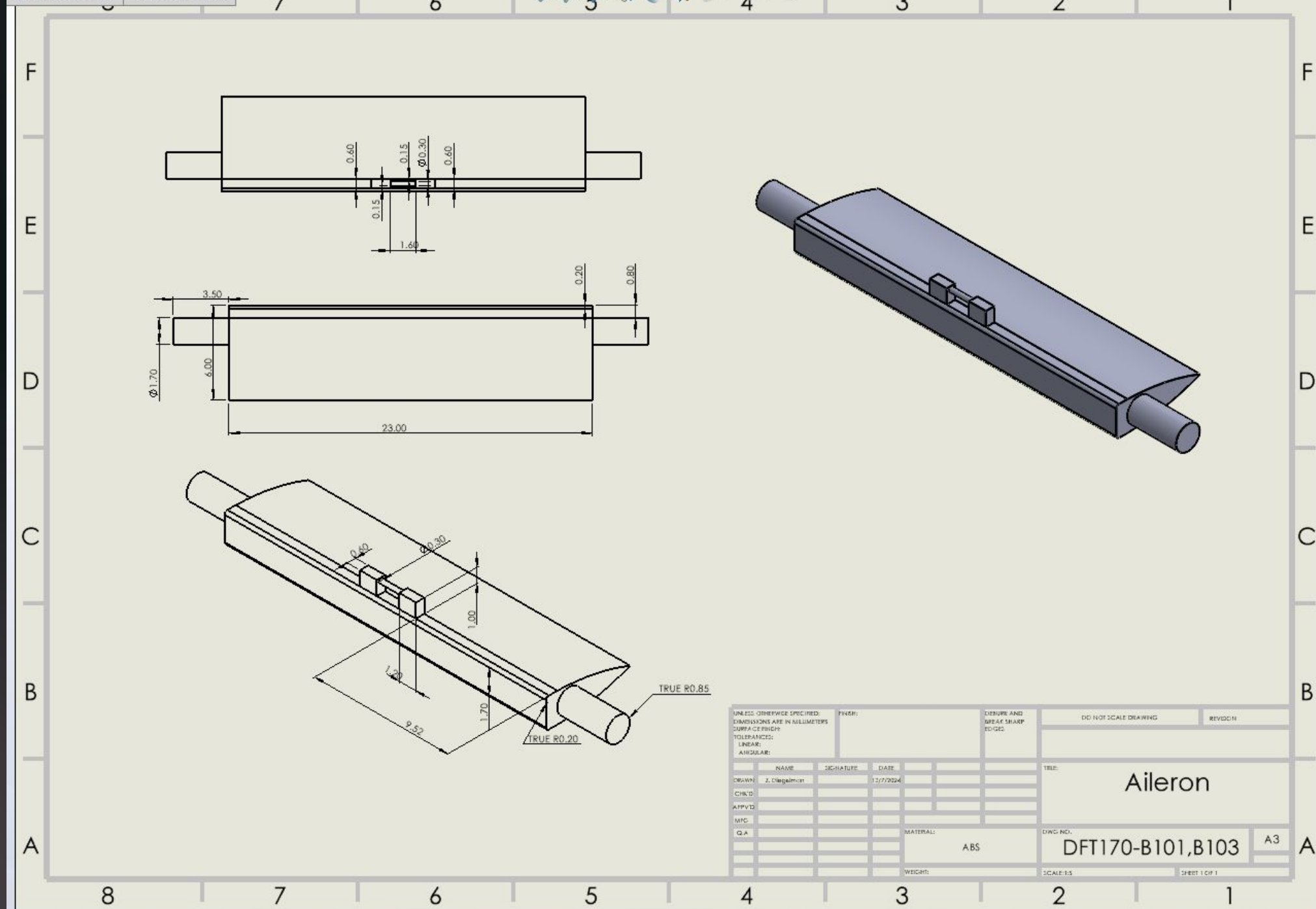




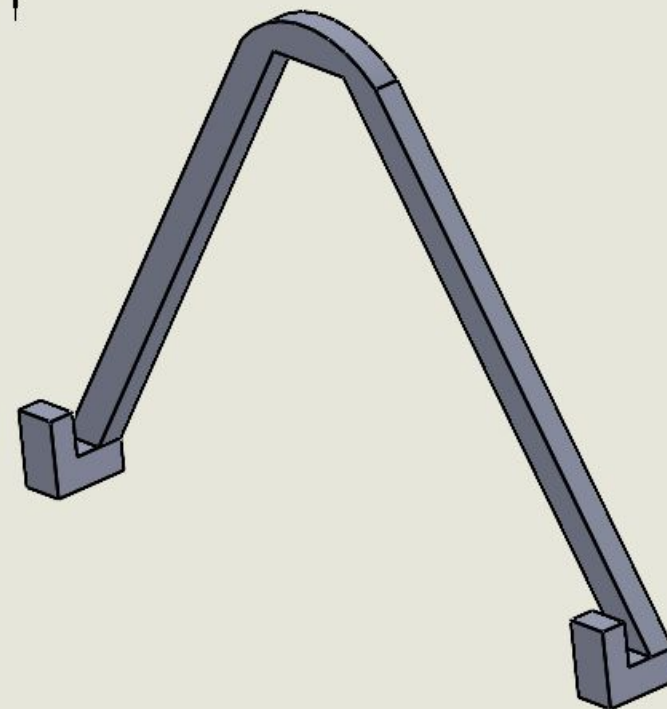
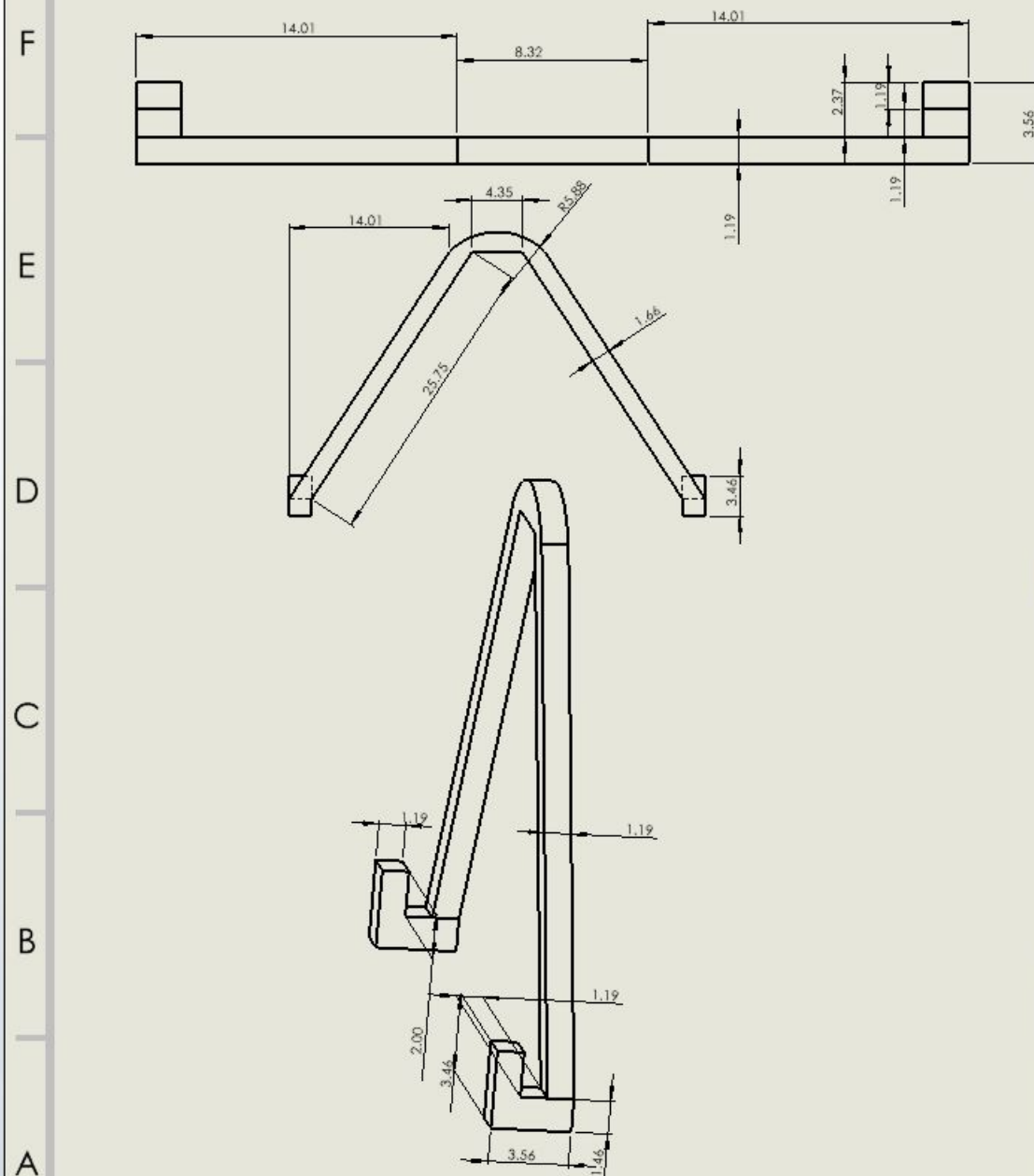
Wing System and Support

- Aileron
- Wings
- Wing Support
- Top Support
- Support Beam Front
- Support Beam Back

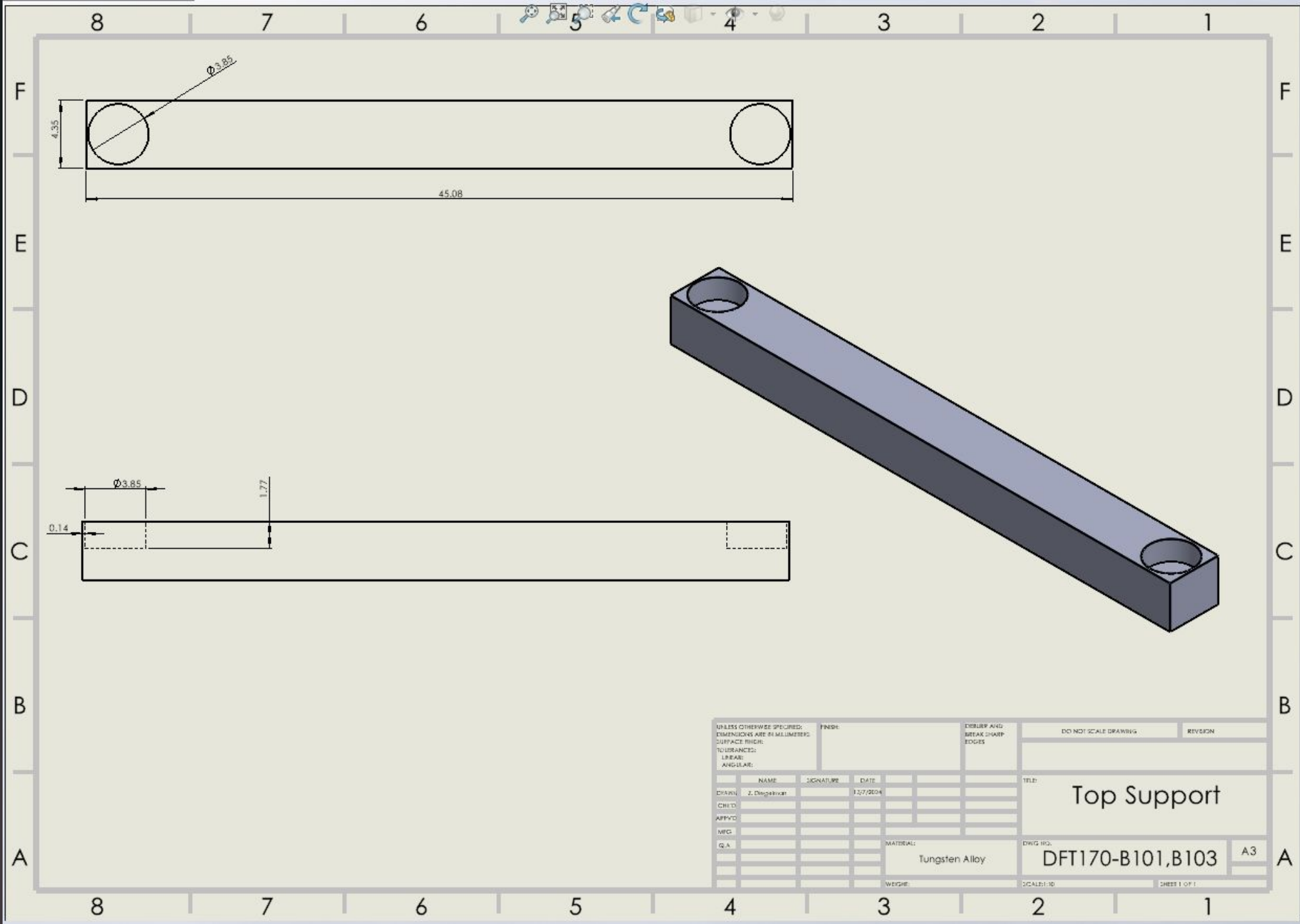


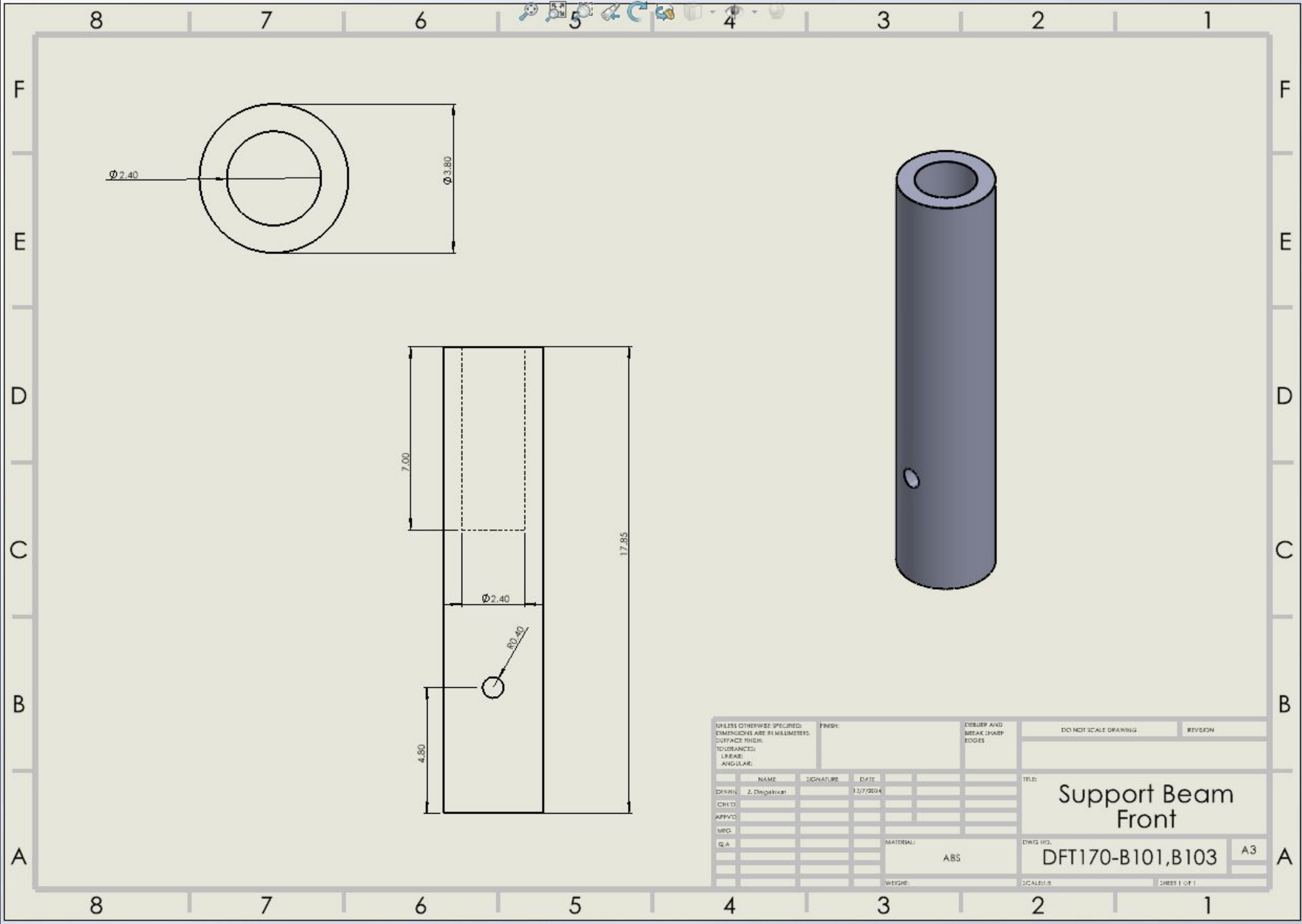


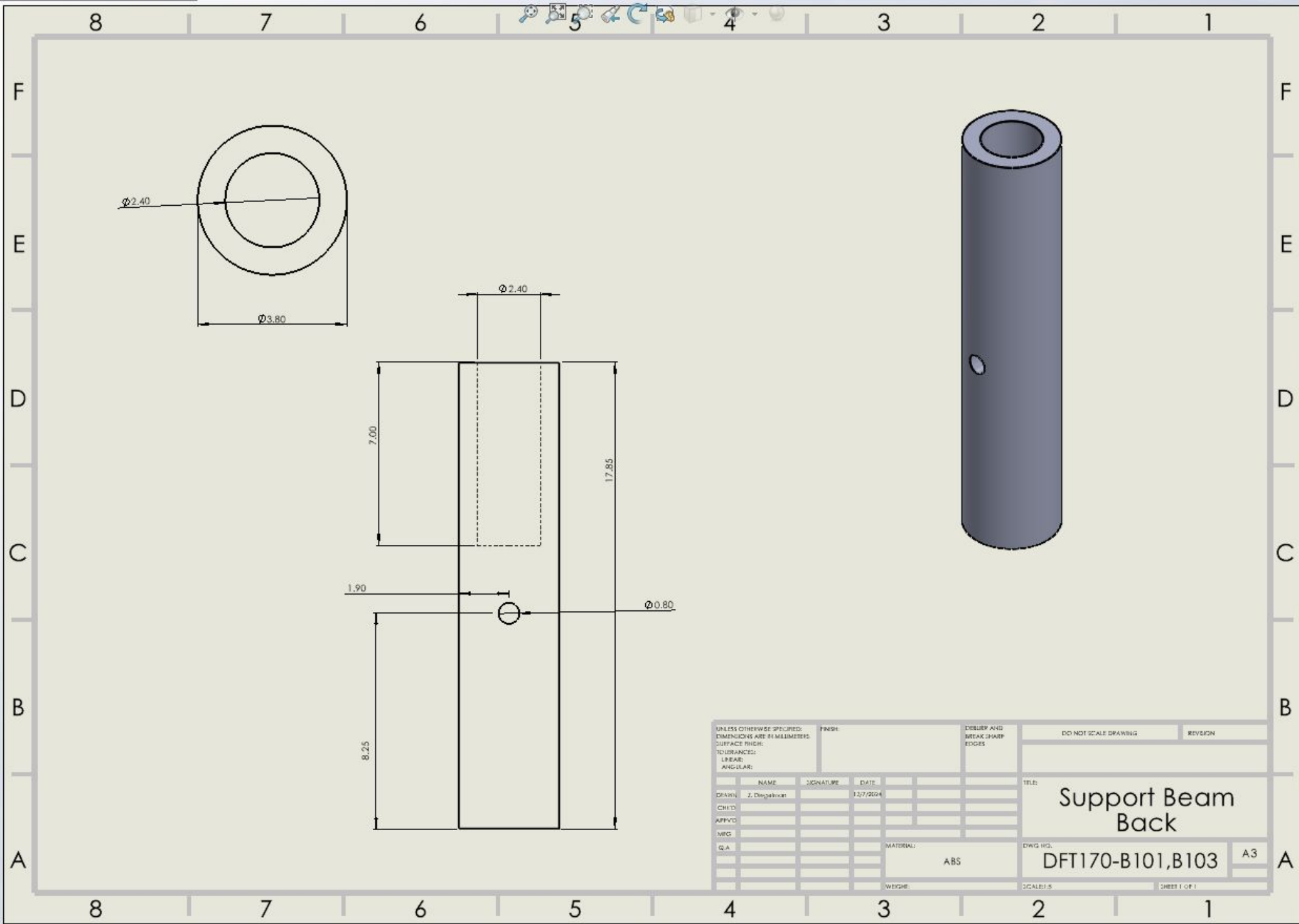
UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: LINEAR: ANGULAR:			PAPER:		DESURE AND BREAK SHARP EDGES		DO NOT SCALE DRAWING		REVISION	
DRAWN: J. Dargatzis			SIGNATURE:		DATE: 11/17/2024		TITLE:		Aileron	
CHK'D:			APPROV'D:		MPC:		QA:		DWC NO.:	
					MATERIAL:		SCALE: 1:1		SHEET 1 OF 1	
					ABS		DFT170-B101,B103		A3	
					WEIGHT:					



UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: LINEAR: ANGULAR:			PAPER:		DEBURE AND GREAT SHARP EDGES		DO NOT SCALE DRAWING		REVISION	
NAME		SIGNATURE		DATE				TITLE:		
DRAWN: J. Dargatzis				12/7/2024				Wing Support		
CHK'D:										
APP'D:										
MFG:										
QA:										







UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: LINEAR: ANGULAR:			FINISH:		DEBurr AND BREAK SHARP EDGES		DO NOT SCALE DRAWING		REVISION	
DRAWN	NAME	SIGNATURE	DATE							
CH'D	Z. Dargatziotis		12/7/2024							
APP'VD										
MFG										
QA										
			MATERIAL:		DWG NO.		SCALE: 1:1		SHEET 1 OF 1	
			ABS		DFT170-B101,B103		A3			
			WEIGHT:							

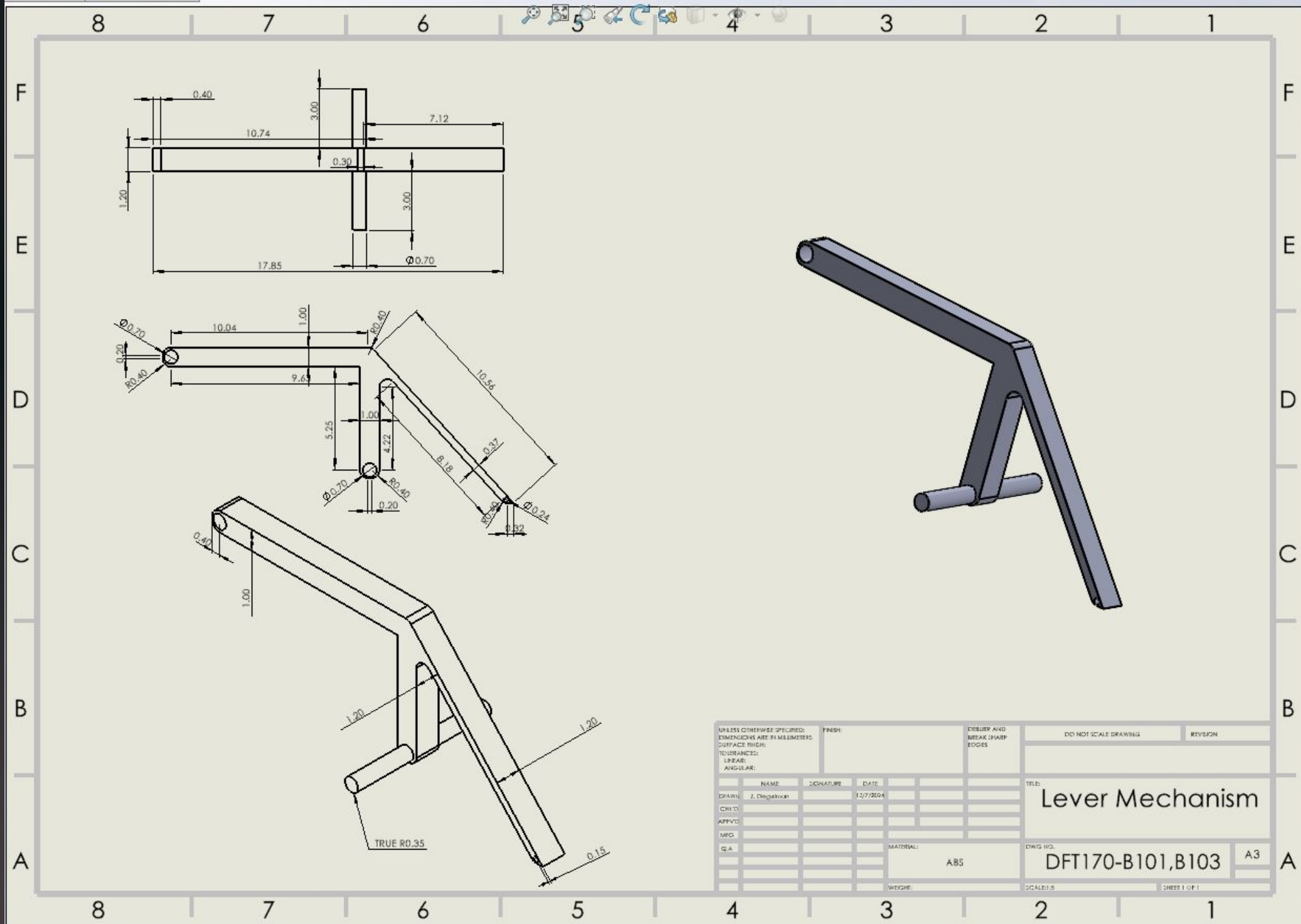
Support Beam
Back

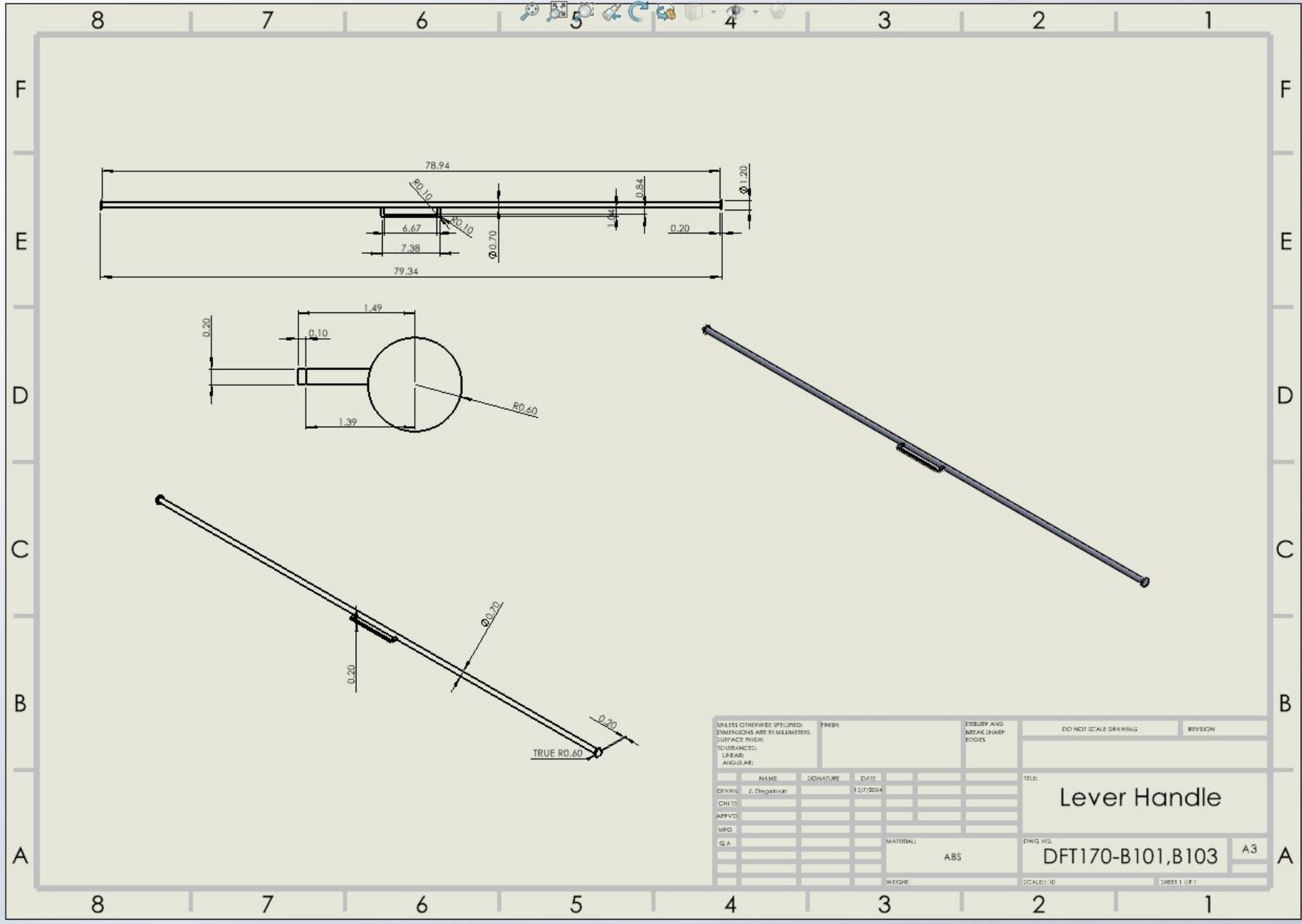
DFT170-B101,B103

A3

Steering System

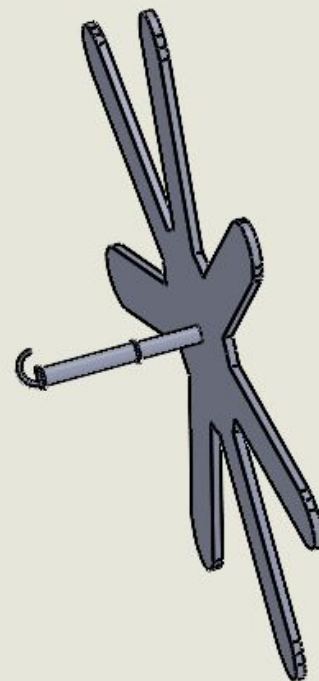
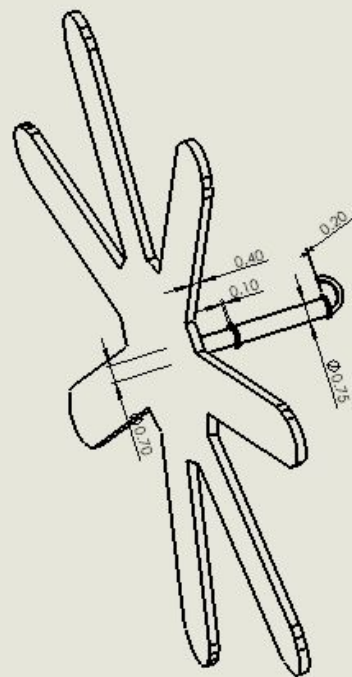
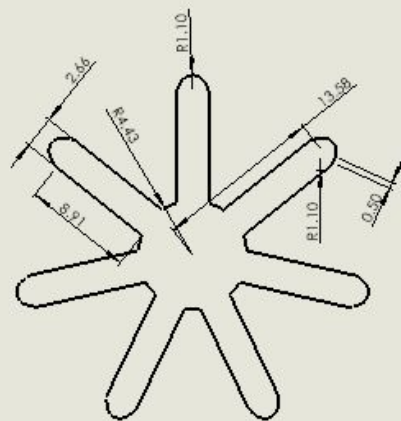
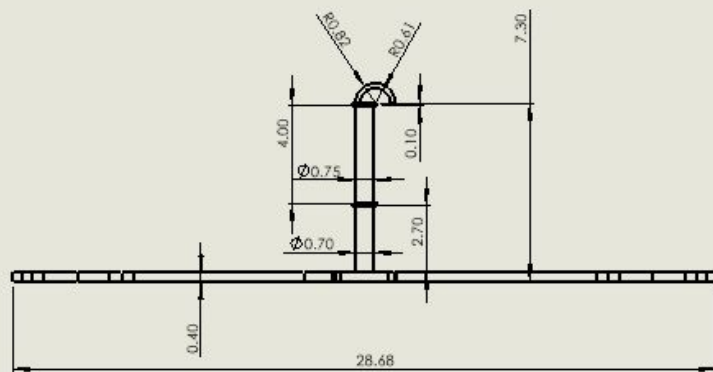
- Lever Mechanism
- Lever Handle





Gear System

- Hook w/ Gear
- Hook w/ Spinning Propeller
- Gear Support Large
- Gear Support Small
- Double Gear Long
- Bevel Gear Large
- Bevel Gear Small
- Rubber Band Prop

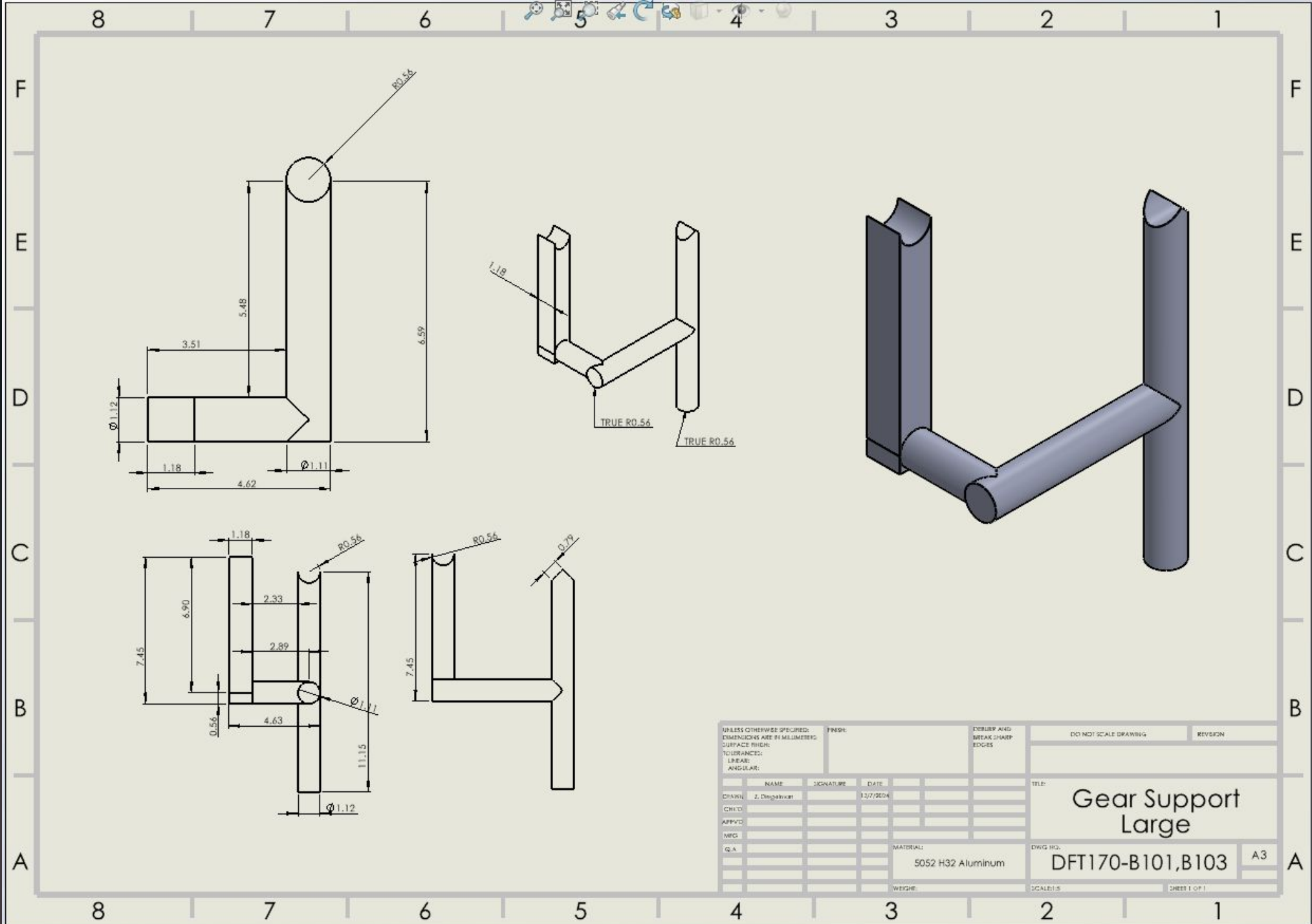


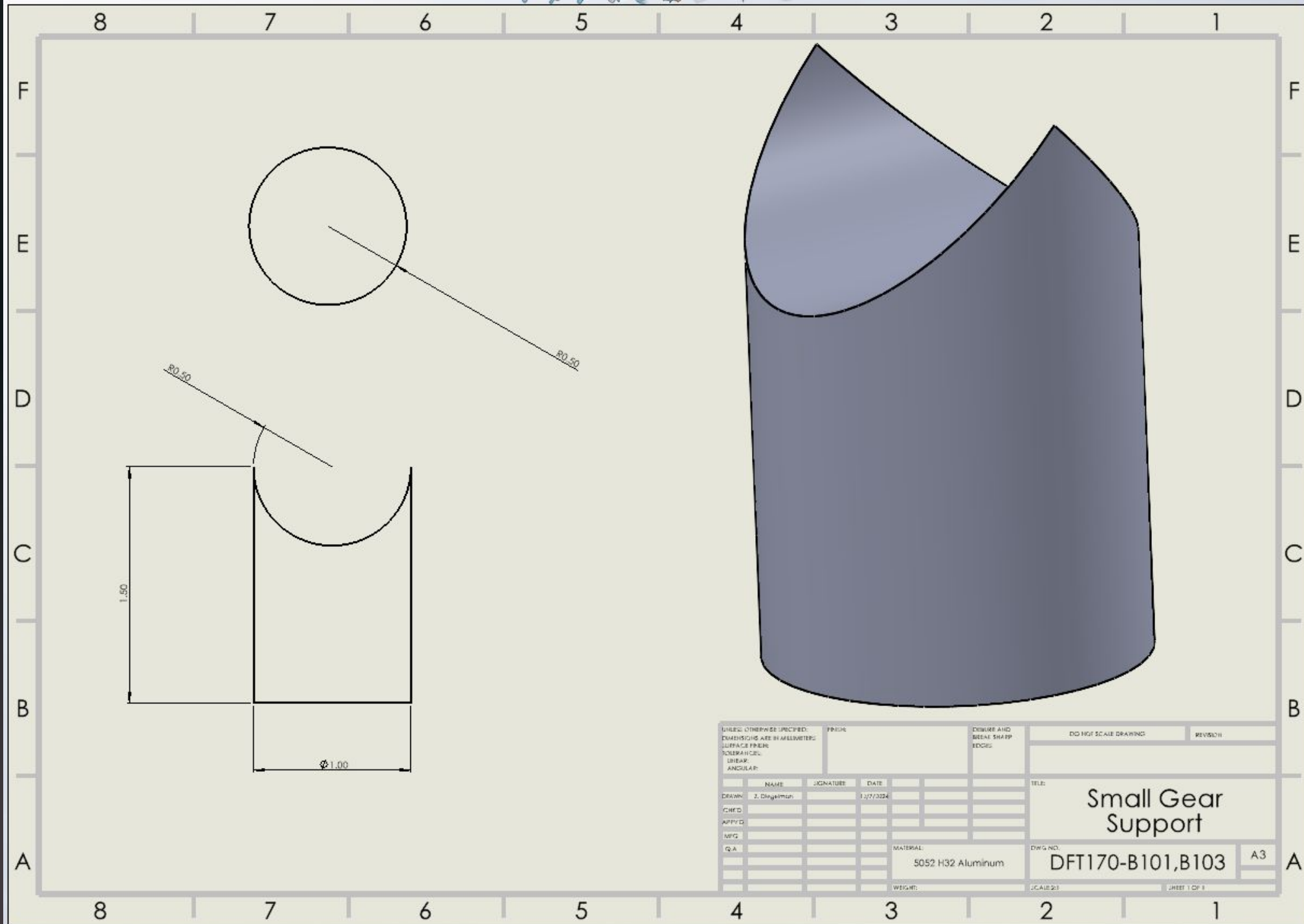
UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS		FINISH:		DRESSER AND BREAK SHARP EDGES		DO NOT SCALE DRAWING		REVISION	
SURFACE FINISH:									
TOLERANCES:									
LINEAR:									
ANGULAR:									
NAME		SIGNATURE		DATE		TITLE			
DRAWN J. Dwyer				12/7/2024		Hook W/ Spinning Propeller			
CHECKED									
APPROVED									
MFG									
Q.A.									
						MATERIAL:		DWG NO.	
						PETG		DFT170-B101,B103	
						WEIGHT:		SCALE: 1:1	
								SHEET 1 OF 1	

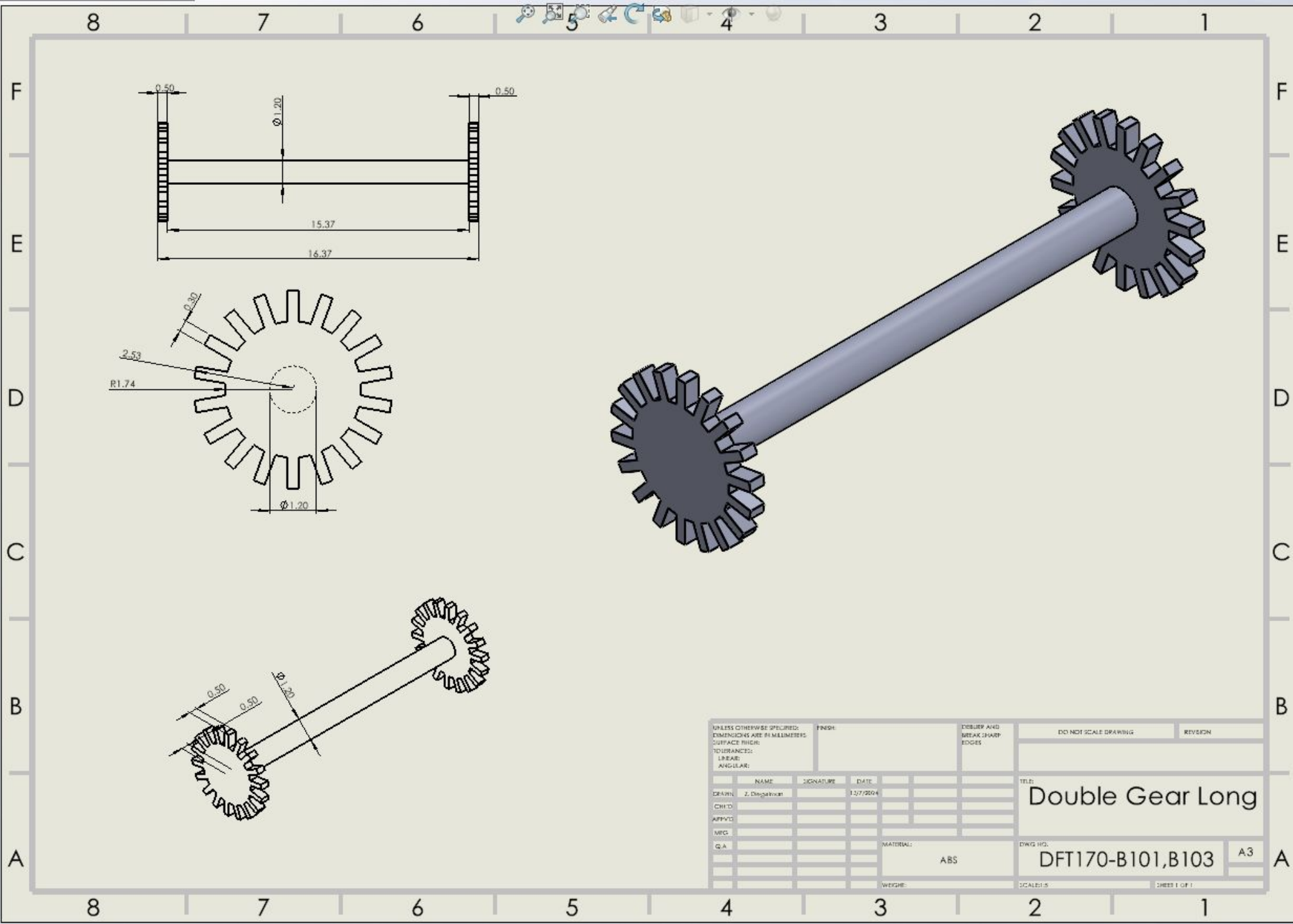
Hook W/ Spinning
Propeller

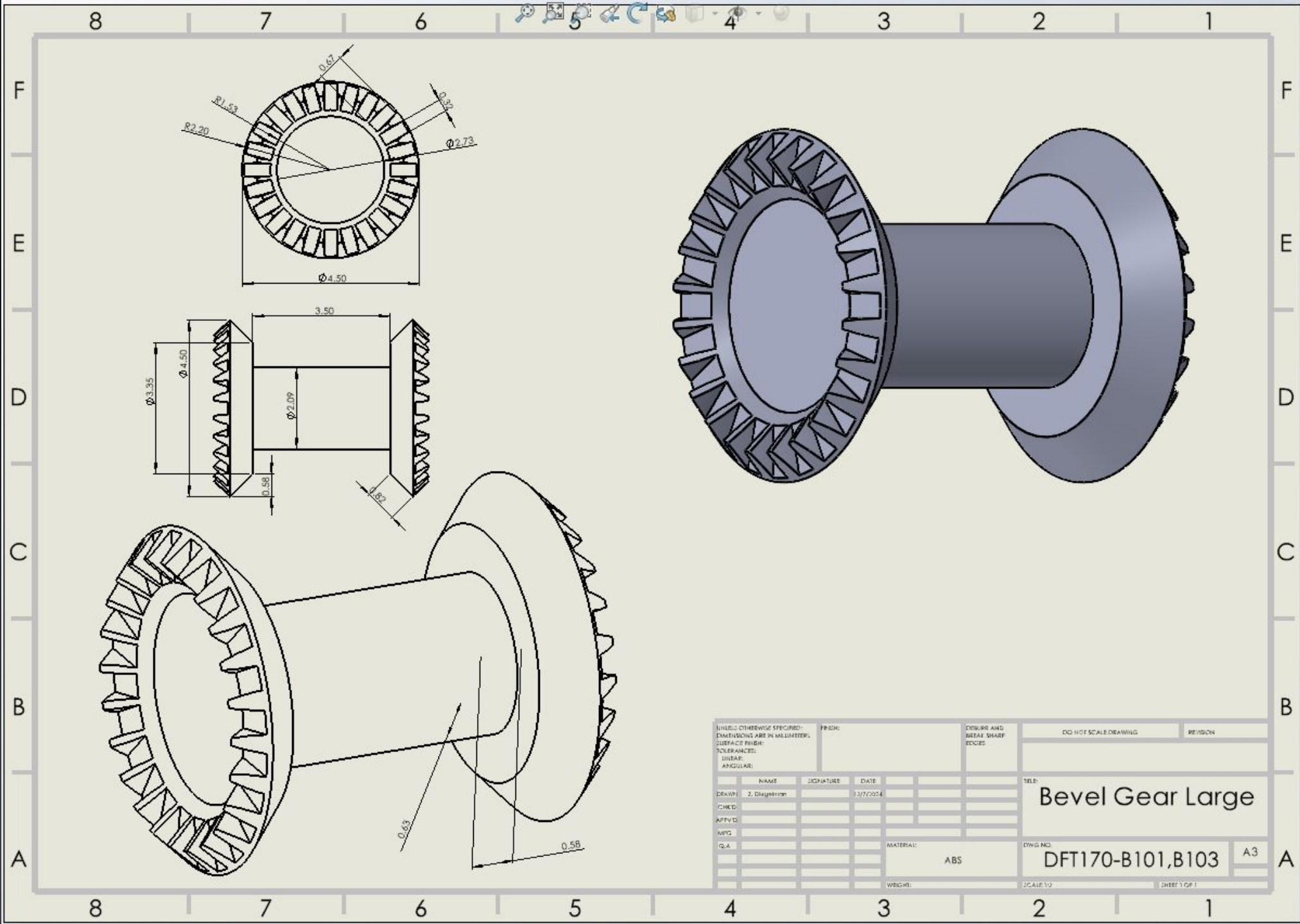
DFT170-B101,B103

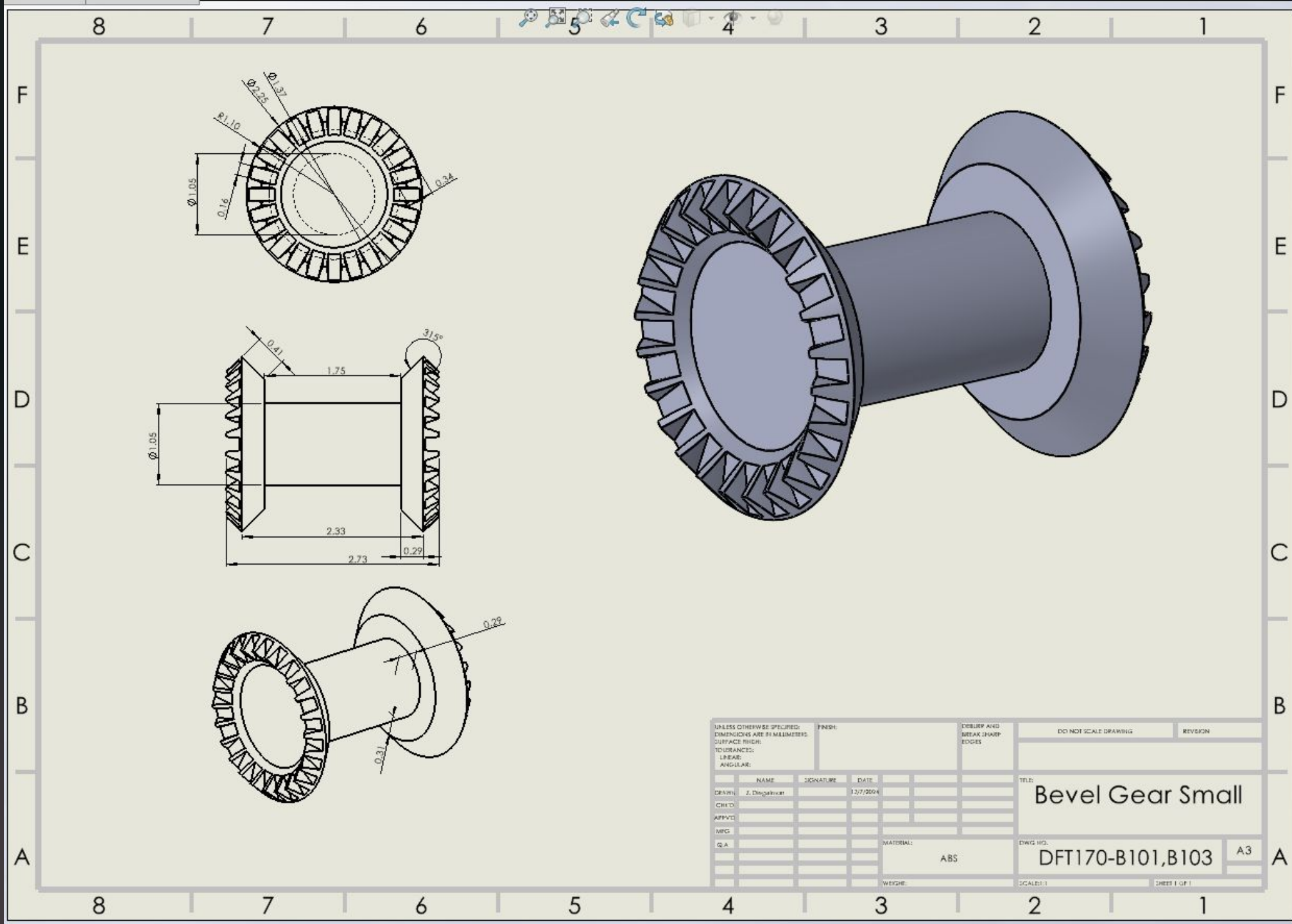
A3



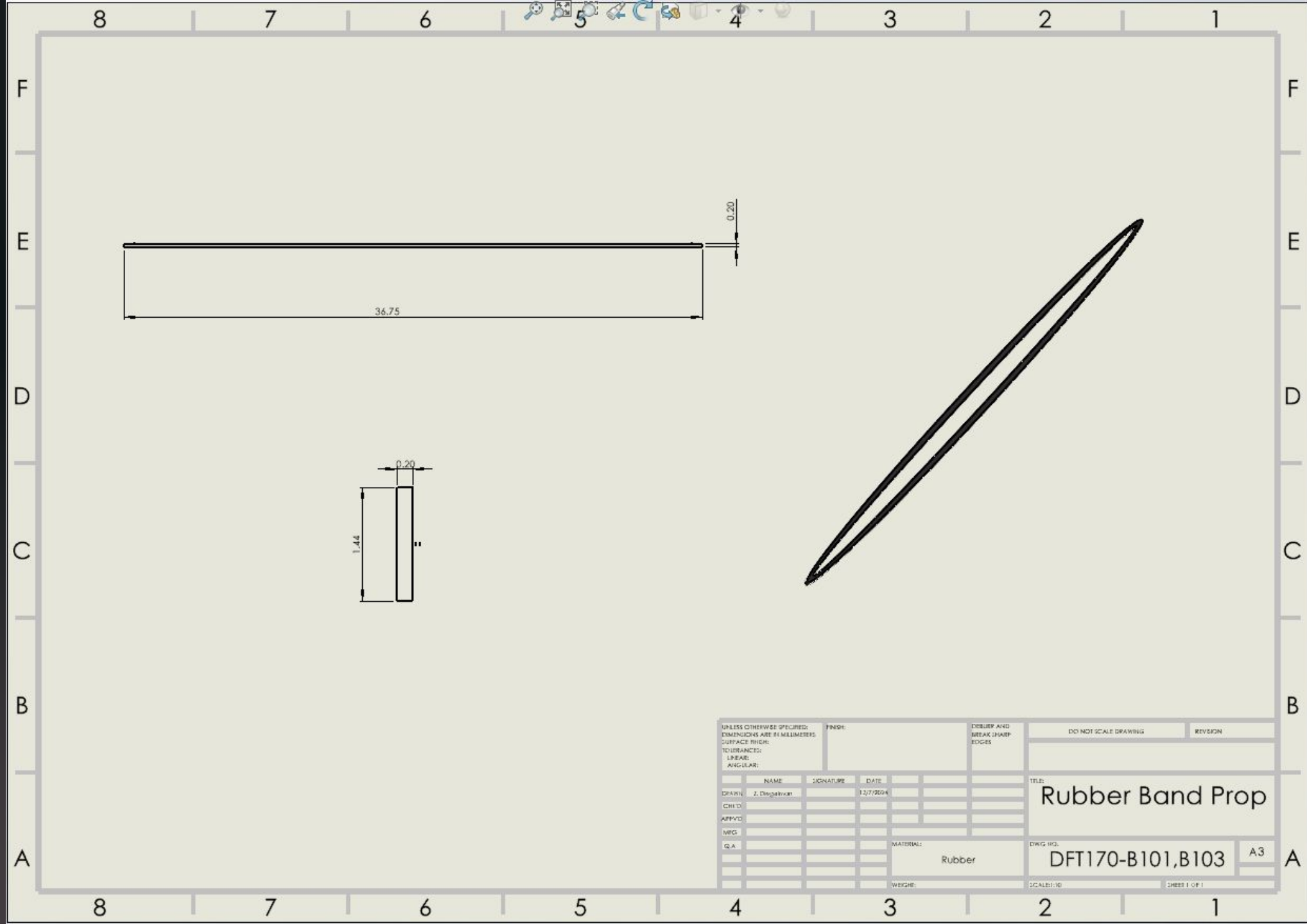








UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: LINEAR: ANGULAR:		FINISH:	DESIGN AND BREAK SHARP EDGES	DO NOT SCALE DRAWING	REVISION
NAME	SIGNATURE	DATE			
DESIGN	J. Dwyer	12/7/2004			
CHIEF					
APPROV					
MPG					
Q.A.					
MATERIAL: ABS			TITLE: Bevel Gear Small		
WEIGHT:			DWG NO. DFT170-B101,B103		
SCALE: 1:1			SHEET 1 OF 1		



Application of Wings & Lever

- The wings are securely fastened within the inner framing of the toy car, supported by beam structures precisely aligned with the measurements of the Carplat. The top support stabilizes the beam supports, while the wing support applies upward pressure on the wing frame and downward force on the top beam support. The wings include a designated area for the lever mechanism to pivot on its axis. The ailerons feature a built-in bar that interlocks with the lever mechanisms, allowing the lever handle to control the plane's upward and downward movements for steering.

Application of the Gears and Twisting Propeller Hooks

- The gears are designed to twist the rubber band propeller as the toy car is pulled back with its wheels on the ground. As the rubber band tightens and builds tension, releasing the car activates the propeller, causing it to spin. The gear supports ensure smooth and continuous gear movement during the pullback. Once the car-plane is airborne, the gears lock, directing all the rubber band's energy to the propeller. The propeller, secured yet free to spin, is positioned within the front support beam. Meanwhile, the back support beam holds the gear with the hook, enabling it to spin during the initial pullback.

Future Recommendations

- Future designs could feature more aerodynamic wings to improve airflow efficiency. Additionally, the top support could be redesigned as a cylindrical shape instead of the current square design, which may hinder the car-plane's aerodynamics. The hook gear could be made smaller, allowing for greater rotation and creating more tension in the rubber band for increased speed and distance. Finally, the propeller blades should be redesigned to optimize airflow, enhancing both velocity and acceleration.

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